



Contents

1	Combinatorics	2
2	Past RMO Papers	26
3	Answer Key	53

Combinatorics

Basic Counting Principles

Addition Principle

When an event can occur in p ways, a second event can occur in q ways, third event can occur in r ways and so on, then any one of these events can occur in $p + q + r + \dots$ ways

Applications

1. In how many ways can the class teacher elect a class president for a class of 15 girls and 20 boys?
2. A stadium is in the shape of an octagon and there is one exit on each side except for the main front which has three exits. In how many ways can a person leave the stadium?
3. There are n locks and n matching keys. If all the locks and keys are to be perfectly matched, find the maximum number of trials required to open all locks.
4. How many 2-digit numbers having different digits, are divisible by 5?

Multiplication Principle

When an event can occur in p ways, a second event can occur in q ways, third event can occur in r ways and so on, then all of these events can occur, in this specified order, in $p \times q \times r \times \dots$ ways.

Applications

1. Every student sent a greeting card to every other student of the class. If the postman delivered 210 cards, find the number of students in the class.
2. 21 people gathered at a party. If every person shook hands, find the total number of handshakes.
3. How many total possible outcomes are there when a coin is tossed n times?
4. How many words, with or without meaning, can be made using all letters of the word "TABLE"?
5. How many even 4-digit numbers are there?
6. How many 5-digit numbers are there of which no two consecutive digits are same?
7. How many natural numbers less than 5000 can be formed using the digits 0, 2, 3, 5, 6?
8. How many 4-digit multiples of 4 can be formed using digits 1,2,3,4,5 without repetition?

9. How many times is the digit 5 written when listing all numbers from 1 to 10^5 ?
10. Find the number of diagonals in a convex n -gon.
11. Find the number of divisors of 129600 which are perfect squares. Find the sum of all these perfect squares.
12. How many integers lie between 1 and 1000 such that $(n, 36) = 1$ (where $(a, b) = \gcd$ of a and b).
13. In how many ways can the following pieces (one white and one black) be placed on a standard chessboard so they do not attack each other?
 - (a) two rooks
 - (b) two kings
14. How many ordered solutions in positive integers (x, y) exist for $\frac{1}{x} + \frac{1}{y} = \frac{1}{4}$? List all the solutions.

Factorials

We define $n!$ as the product of first n natural numbers for all $n \in \mathbb{N}$. The factorial is related to number of arrangements of n distinct objects in n distinct boxes. By multiplication principle, this evaluates as

$$n! = n \times (n-1) \times (n-2) \times \dots \times 2 \times 1.$$

A recursive relation that follows from above : $(n+1)! = n! \times (n+1)$

$0!$ is defined as 1. Since, there is only one way of arranging zero objects, namely, arrange nothing.

Applications

1. $\frac{1}{8!} + \frac{1}{9!} = \frac{x}{10!}$. Find x .
2. Determine the units digit of $1! + 2! + 3! + \dots + 20!$.
3. Find the remainder when $1! + 2! + 3! + \dots + 15!$ is divided by 15.
4. Simplify : $1 \cdot 1! + 2 \cdot 2! + 3 \cdot 3! + \dots + 49 \cdot 49!$.
5. Compute the units digit of $(1!)^{20!} + (2!)^{19!} + \dots + (20!)^{1!}$.
6. Someone observed that $6! = 8 \cdot 9 \cdot 10$. Find the largest positive integer n for which $n!$ can be expressed as the product of $n-3$ consecutive positive integers.

Permutations

Distinct Permutations

An r - *permutation* means taking r objects out of n distinct objects and arranging them in r distinct boxes. The number of ways an r -permutation of n objects can take place is denoted by ${}^n P_r$. By multiplication principle,

$${}^n P_r = n \times (n-1) \times (n-2) \times \dots \times (n-(r-1)) = \frac{n!}{(n-r)!}$$

Applications

1. In how many ways can 4 Latin and 1 English book be placed on a shelf so that the English book is always in the middle, the selection being made from 7 Latin and 3 English books?
2. In how many arrangements, do the girls and boys alternate when
 - (a) 5 girls and 6 boys
 - (b) 6 girls and 6 boysstand in a line?
3. There are 7 boys and 3 girls in a gathering. In how many ways can they be arranged in a row so that
 - (a) the 3 girls stand together?
 - (b) boys occupy end-positions and no girls are adjacent?
4. In how many permutations of the word EQUATION do the vowels occur in alphabetical order? For how many permutations the relative positioning of vowels and consonants does not change?
5. In how many ways can 10 examination papers be arranged so that the best and worst papers never come together?
6. There are two works each of 3 volumes and two works each of 2 volumes. In how many ways can the 10 books be placed on a shelf so that volumes of same work are not separated?
7. A, B, C, D and E are five speakers who have to address a crowd. In how many ways can
 - (a) A speak immediately after B?
 - (b) A speak after B?
8. Find the sum of all 5-digit numbers that can be formed by taking digits from $\{1,2,3,4,5\}$ once each.

Permutations with Repetitions

When out of n total objects, p objects are alike, q objects are alike, r objects are alike and rest are all different, the distinct number of permutations is given by

$$\frac{n!}{p!q!r!}$$

Note : One can start with n distinct objects, permuting them in $n!$ ways. Since the p distinct objects can be permuted among themselves in $p!$ ways, when they are all identical, one divides by $p!$ to correct the count of all permutations only among the identical objects, as only once, and so on.

Applications

1. In how many ways can 17 billiards balls be arranged in a row if 7 of them are black, 6 red and 4 white?
2. How many numbers greater than a million can be formed with the digits 2, 3, 0, 3, 4, 2, 3?
3. How many 5-lettered palindromes can be written using 26 letters of English language?
4. All distinct permutations of the word “BATTLE” are written down as in a dictionary.
 - (a) How many words are there in this dictionary?
 - (b) Write down the first and last words.
 - (c) What is the rank of the word “TABLET”?
 - (d) Which word is ranked 300th?
5. How many anagrams of “MISSISSIPPI” (including “MISSISSIPPI”) are there? In how many of these anagrams are no two S’s adjacent?
6. Blocks of sizes 1×1 , 1×2 and 1×3 are available to pave a 1×5 rectangular block. Assuming that the blocks of same size are indistinguishable, find the number of ways to pave the rectangle.

Circular Permutations

A circular permutation differs from a linear permutation as it lacks a starting point to initiate counting. Many permutations in a row such as ABCD, BCDA, CBDA and DABC become identical when put around a circle.

We thus “fix” any one element from n distinct objects and count relative to this “fixed” element. The remaining $(n-1)$ distinct objects can be permuted in $(n-1)!$ ways, which has counted all the circular permutations of n distinct objects as well.

Applications

1. How many different necklaces can be strung from 8 beads of different colors?
2. In how many ways can 5 boys and 3 girls be seated around a table if
 - (a) there is no restriction?
 - (b) boy B1 and girl G1 are not adjacent?
 - (c) no girls are adjacent?
3. Find the number of ways to seat 7 married couples around a table in each of the following cases :
 - (a) men and women alternate?
 - (b) every woman is next to her husband?
4. If there must be at least one person at each table, in how many ways can five people be seated around two identical round tables?
5. How many necklaces can be made using 7 beads of which 5 are identical red beads and 2 are identical blue beads?

6. In how many ways can 5 boys and 16 girls be arranged around a circular table so that there are atleast two girls between any two boys?
7. Six faces of a cube have to be painted with six different colors. How many distinct colorings are possible? Two colorings are distinct when one cannot be obtained from the other by rotation.
8. A standard die of ludo game has the numbers on opposite faces, all summing to 7. Suppose the numbers from 1 to 6 can be randomly marked on each face of a die. How many dice are there which are different from a standard die?

Combinations

r-Combinations

An r – *combination* means selecting r distinct objects out of n distinct objects. The number of ways an r -combination of n objects can take place is denoted by nC_r or $\binom{n}{r}$. $\binom{n}{r}$ is read as “ n choose r ”.

Selecting r distinct objects out of n distinct objects can be done in nC_r ways and arranging these can be done in $r!$ ways. This is same as r -permutation of n . So ${}^nC_r \times r! = {}^nP_r \Rightarrow {}^nC_r = \frac{n!}{r!(n-r)!}$

Applications

1. One student has 6 Maths books and another has 8 books. How many ways are there to exchange 3 books belonging to the first with 3 books belonging to the second?
2. There are 2 girls and 7 boys in a chess club. Find number of ways to make a team of five when there must be atleast 1 girl on the team?
3. How many ways to divide 10 boys into two teams?
4. There are 30 members in a Sports Club, and the club's coach must choose 4 men for a 1000m run.
 - (a) How many ways are there to do this?
 - (b) How many ways to choose a team of 4 members for the 100m + 200m + 300m + 400m relay?
5. How many chords can be drawn through 8 points lying on a circle? Also find the maximum number of points in which these chords can intersect.
6. How many (a) triangles (b) quadrilaterals can be drawn in a convex 20-gon with their vertices on the vertices of the polygon?
7. There are 13 points in a plane, 5 of which are collinear. How many distinct straight lines can be drawn through these points?
8. Fifteen distinct points are designated on $\triangle ABC$: the 3 vertices A, B, and C; 3 other points on side $\overline{AB}$; 4 other points on side $\overline{BC}$; and 5 other points on side $\overline{CA}$. Find the number of triangles with positive area whose vertices are among these 15 points.
9. In how many ways can four people be selected from a group of nine, when
 - (a) two particular people have to be selected together.

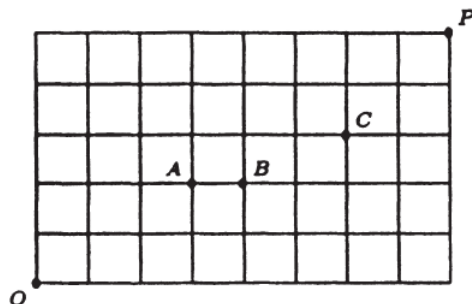
- (b) two particular people do not want to be on same group.
10. How many (a) squares (b) rectangles are there on an 8×8 chessboard?
 11. How many ways to form following shapes from the vertices of a cube?
 - (a) an equilateral triangle
 - (b) an isosceles triangle
 - (c) a right triangle
 - (d) a scalene triangle
 - (e) a square
 - (f) a rectangle
 12. 27 points are marked at uniform distances on the circumference of a circle. How many (a) equilateral (b) scalene triangles can be formed?
 13. Three cards are selected at random from a standard deck of 52 cards. In how many ways can one select the cards such that (a) all of them are from the same suit? (b) all of them are from different suits?
 14. How many full houses (three-of-a-kind and a pair) are there?
 15. The results of 20 chess games (win, lose or draw) have to be predicted. How many different forecasts will contain exactly 15 correct results?
 16. How many ways are there to place 2 identical rooks in a common row or column of an 8×8 chessboard?
 17. Find the number of ways of arranging seven 1's and seven 0's in a line such that no two 1's are adjacent.
 18. Find the number of permutations $(i_1, i_2, \dots, i_{10})$ of the ten digits $\{0, 1, \dots, 9\}$ in which
 - (a) $i_1 < i_2 < i_3$
 - (b) $i_1 < i_2$ and $i_3 < i_4$
 19. Consider the collection of all 3-element subsets drawn from the set $\{1, 2, 3, \dots, 99, 100\}$. Determine the number of subsets for which the sum of the three elements is a multiple of 3.
 20. How many 3-element subsets of the set $\{1, 2, 3, \dots, 20\}$ are there such that the product of three numbers in the subset is divisible by 4?
 21. A four-digit number $abcd$ is called containing if the number ad is bigger than bc . (For example, 4291 and 4091 are containing numbers, but 2359 is not a containing number.) Find the number of containing numbers.

Lattice Points and Paths

A “*lattice point*” is a point in Cartesian plane which both coordinates are integers. A “*lattice path*” is a path in infinite Cartesian plane, which is always parallel to either x-axis or y-axis and has ‘turns’ only at lattice points. A lattice path can have any length, it may be infinite also. The total number of lattice paths between two given points on Cartesian plane is also infinite. However, there are only finitely many lattice paths of a given length between two specific points.

Applications

- Find the number of lattice points enclosed within the triangle formed by coordinate axes and the line $x + y = 17$.
- A bug starts crawling from origin $(0,0)$. Find the number of shortest lattice paths it can take to reach $(10,10)$
 - under no restrictions
 - via $(3,2)$
 - via $(6,8)$
 - via $(3,2)$ as well as $(6,8)$
 - through $(3,2)$ but not through $(6,8)$
 - neither through $(3,2)$ nor through $(6,8)$
- In the street network given below, find the number of shortest routes from O to P in each of following cases:
 - the routes must pass through the street AB.



- the street AB is closed.
- Find the number of lattice paths from $(2,3)$ to $(11,14)$ of length (a) 20 units (b) 24 units.

Number of Integral Solutions

Consider the equation

$$x_1 + x_2 + \dots + x_r = n$$

where each $x_i \in \mathbb{Z}, x_i \geq 1$.

Any solution corresponds to putting $r-1$ '+' signs in distinct gaps between n '1's. Hence the total number of positive integral solutions to the above equation is ${}^{n-1}C_{r-1}$.

For the following equation

$$y_1 + y_2 + \dots + y_r = n$$

where each $y_i \in \mathbb{Z}, y_i \geq 0$. So every y_i is a non-negative integer.

The total number of non-negative solutions to above equation is ${}^{n+r-1}C_{r-1}$ or ${}^{n+r-1}C_n$.

Applications

1. Find the number of solutions in non-negative integers for $4x + y + z = 10$.
2. Find the number of integral solutions for $x_1 + x_2 + x_3 + x_4 = 10$ where $x_1 \geq -2$, $x_2 > -5$, $x_3 \geq 0$, $x_4 > 4$.
3. In how many ways can one choose 6 candies from 8 brands that are available? How many ways for 9 candies?
4. A 6-number sequence is said to be “*non – decreasing*” if the digits never decrease while reading from left to right eg, 223337. Find the number of all non-decreasing 6-sequences.
5. $x_1 + x_2 + x_3 + x_4 = 10$ has same number of positive integral solutions as $x_1 + x_2 + \dots + x_m = 10$ ($m \neq 4$). Find the value of m .
6. Find total number of positive integral solutions of $p + q + r + s < 15$.
7. Given eight distinguishable rings, find the number of possible five-ring arrangements on the four fingers (not on the thumb) of one hand. (The order of the rings on each finger is not significant, neither it is required that each finger have a ring).
8. A grasshopper stands on the leftmost box of a 16×1 strip of squares and can jump any number of boxes to the right at a time.
 - (a) How many ways are there for it to reach the rightmost box?
 - (b) How many ways are there to reach the rightmost box in exactly four moves?

Binomial Theorem

For any $n \geq 0$, $(a + b)^n = \sum_{r=0}^n {}^nC_r a^{n-r} b^r = \sum \binom{n}{r} a^{n-r} b^r$,
where $\binom{n}{r} = \frac{n!}{r!(n-r)!}$.

- Each term in the expansion of $(a + b)^n$ has degree n . Hence, the total number of terms is $n + 1$.
- The coefficient of $a^{n-r} b^r$ is nC_r or $\binom{n}{r}$. $\binom{n}{r}$ are known as *binomial coefficients*.
- Coefficients of $a^{n-r} b^r$ and $a^r b^{n-r}$ are equal, $\binom{n}{r} = \binom{n}{n-r}$.
- Putting $a = 1, b = x$, $(1 + x)^n = \sum_{r=0}^n \binom{n}{r} a^{n-r} b^r$.

Combinatorial Identities

Applications

1. Expand : $(x + 2y)^4$.
2. $(3p - q)^7 = a_7 p^7 + a_6 p^6 q + \dots + a_1 p q^6 + a_0 q^7$. Find $(a_7 + a_6 + \dots + a_0)$ and $(a_7 - a_6 + a_5 - a_4 + \dots - a_0)$.
3. Find the constant term in $(x - \frac{1}{x})^6$.
4. Expand each of $[(x + y)^5 + (x - y)^5]$ and $[(x + y)^5 - (x - y)^5]$.

5. Prove the following sums :

- (a) $\sum^n C_r = 2^n$
- (b) ${}^nC_0 - {}^nC_1 + {}^nC_2 - {}^nC_3 + \dots + (-1)^n ({}^nC_n) = 0$
- (c) ${}^nC_1 + {}^nC_3 + {}^nC_5 + \dots = {}^nC_0 + {}^nC_2 + {}^nC_4 + \dots = 2^{n-1}$
- (d) $({}^nC_0)^2 + ({}^nC_1)^2 + \dots + ({}^nC_n)^2 = 2^n {}^nC_n$

6. Provide combinatorial arguments for each of the following :

- (a) ${}^{n+1}C_r = {}^nC_r + {}^nC_{r-1}$
- (b) ${}^nC_r = {}^nC_{n-r} \Leftrightarrow {}^{m+n}C_m = {}^{m+n}C_n$
- (c) $\sum^n C_r = 2^n$
- (d) $1 + 2 + 3 + \dots + n = \binom{n+1}{2}$
- (e) $\binom{n}{0}^2 + \binom{n}{1}^2 + \dots + \binom{n}{n}^2 = \binom{2n}{n}$
- (f) $\binom{m}{0}\binom{n}{k} + \binom{m}{1}\binom{n}{k-1} + \binom{m}{2}\binom{n}{k-2} + \dots + \binom{m}{k}\binom{n}{0} = \binom{m+n}{k}$ (Vandermonte's Identity)
- (g) $r^n C_r = n^{n-1} C_{r-1}$

7. Show that there is only an even number of ways to choose n objects from a total of $2n$ objects.

8. Evaluate : $\binom{8}{8} + \binom{9}{8} + \binom{10}{8} + \binom{11}{8} + \binom{12}{8} + \binom{13}{8} + \binom{14}{8}$

9. Find the number of 10-digit binary sequences (leading zeros allowed) in which the number of 0's in the first 5 digits is equal to the number of 1's in the last 5 digits.

10. Use a combinatorial argument to prove that $\sum_{k=i}^n \binom{k}{i} = \binom{n+1}{i+1}$ for $n \geq 0$ and $i \geq 0$; that is, explain why the left-hand side counts the same thing as the right-hand side.

11. Use a combinatorial argument to prove that $\binom{k}{2} + \binom{n-k}{2} + k(n-k) = \binom{n}{2}$.

12. Define a sequence $a_n = 6^n + 8^n$. Find the remainder when a_{83} is divided by 49.

Pascal's Triangle

The binomial coefficients arranged in a triangular array, make up what is known as Pascal's triangle.

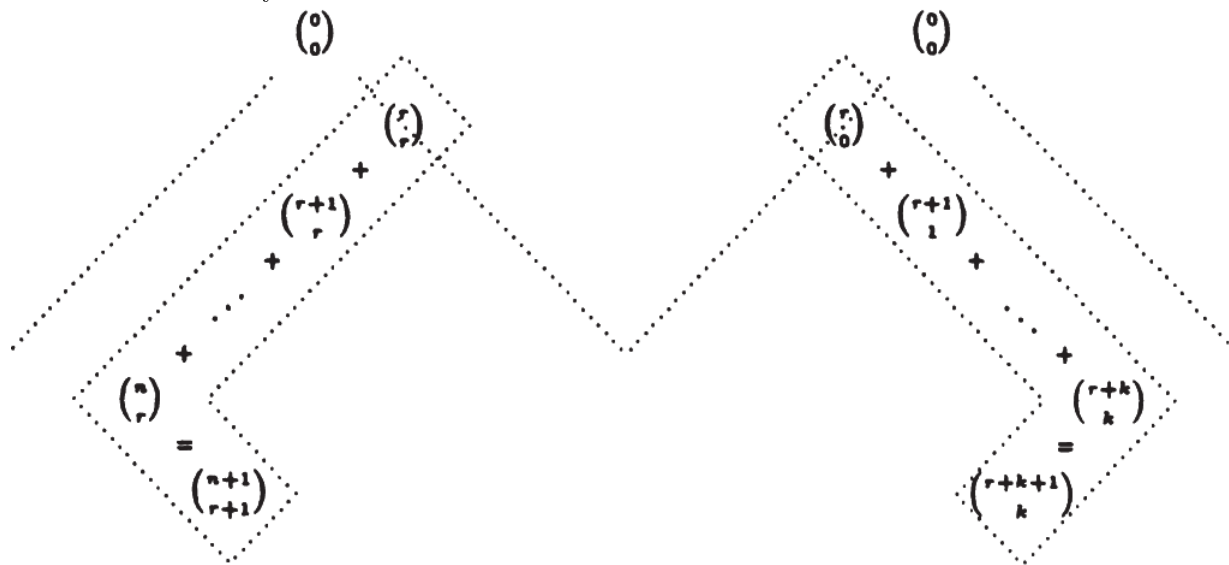
$$\begin{array}{ccccccc}
 & & & & & & \binom{0}{0} \\
 & & & & & & \\
 & & & & & & \binom{1}{0} & \binom{1}{1} \\
 & & & & & & \\
 & & & & & & \binom{2}{0} & \binom{2}{1} & \binom{2}{2} \\
 & & & & & & \\
 & & & & & & \binom{3}{0} & \binom{3}{1} & \binom{3}{2} & \binom{3}{3} \\
 & & & & & & \\
 & & & & & & \binom{4}{0} & \binom{4}{1} & \binom{4}{2} & \binom{4}{3} & \binom{4}{4}
 \end{array}$$

Each row of Pascal's triangle has 1 at the beginning as well as at end. Any element in a row is found by adding two adjacent elements in the previous row. This rule is derived from the $\binom{n+1}{r} = \binom{n}{r-1} + \binom{n}{r}$ property

of binomial coefficients.

					1				
				1		1			
			1		2		1		
		1		3		3		1	
	1		4		6		4		1
	1	5		10		10		5	
1		6		15		20		15	
1	7		21		35		35		21
1	7	21		35		35		21	
1	7	21		35		35		21	
1	7	21		35		35		21	
1	7	21		35		35		21	
1	7	21		35		35		21	
1	7	21		35		35		21	
1	7	21		35		35		21	
1	7	21		35		35		21	
1	7	21		35		35		21	
1	7	21		35		35		21	
1	7	21		35		35		21	
1	7	21		35		35		21	
1	7	21		35		35		21	
1	7	21		35		35		21	
1	7	21		35		35		21	
1	7	21		35		35		21	
1	7	21		35		35		21	
1	7	21		35		35		21	
1	7	21		35		35		21	
1	7	21		35		35		21	
1	7	21		35		35		21	
1	7	21		35		35		21	
1	7	21		35		35		21	
1	7	21		35		35		21	
1	7	21		35		35		21	
1	7	21		35		35		21	
1	7	21		35		35		21	
1	7	21		35		35		21	
1	7	21		35		35		21	
1	7	21		35		35		21	
1	7	21		35		35		21	
1	7	21		35		35		21	
1	7	21		35		35		21	
1	7	21		35		35		21	
1	7	21		35		35		21	
1	7	21		35		35		21	
1	7	21		35		35		21	
1	7	21		35		35		21	
1	7	21		35		35		21	
1	7	21		35		35		21	
1	7	21		35		35		21	
1	7	21		35		35		21	
1	7	21		35		35		21	
1	7	21		35		35		21	
1	7	21		35		35		21	
1	7	21		35		35		21	
1	7	21		35		35		21	
1	7	21		35		35		21	
1	7	21		35		35		21	
1	7	21		35		35		21	
1	7	21		35		35		21	
1	7	21		35		35		21	
1	7	21		35		35		21	
1	7	21		35		35		21	
1	7	21		35		35		21	
1	7	21		35		35		21	
1	7	21		35		35		21	
1	7	21		35		35		21	
1	7	21		35		35		21	
1	7	21		35		35		21	
1	7	21		35		35		21	
1	7	21		35		35		21	
1	7	21		35		35		21	
1	7	21		35		35		21	
1	7	21		35		35		21	
1	7	21		35		35		21	
1	7	21		35		35		21	
1	7	21		35		35		21	
1	7	21		35		35		21	
1	7	21		35		35		21	
1	7	21		35		35		21	
1	7	21		35		35		21	
1	7	21		35		35		21	
1	7	21		35		35		21	
1	7	21		35		35		21	
1	7	21		35		35		21	
1	7	21		35		35		21	
1	7	21		35		35		21	
1	7	21		35		35		21	
1	7	21		35		35		21	
1	7	21		35		35		21	
1	7	21		35		35		21	
1	7	21		35		35		21	
1	7	21		35		35		21	
1	7	21		35		35		21	
1	7	21		35		35		21	
1	7	21		35		35		21	
1	7	21		35		35		21	
1	7	21		35		35		21	
1	7	21		35		35		21	
1	7	21		35		35		21	
1	7	21		35		35		21	
1	7	21		35		35		21	
1	7	21		35		35		21	
1	7	21		35		35		21	
1	7	21		35		35		21	
1	7	21		35		35		21	
1	7	21		35		35		21	
1	7	21		35		35		21	
1	7	21		35		35		21	
1	7	21		35		35		21	
1	7	21		35		35		21	
1	7	21		35		35		21	
1	7	21		35		35		21	
1	7	21		35		35		21	
1	7	21							

- The sum of all entries in n^{th} row of Pascal's triangle is 2^n .
- The numbers in n^{th} row are symmetric about the middle term(s). The middle term of an even row is $\binom{n}{n/2}$. The two middle terms of an odd row are $\binom{n}{(n-1)/2} = \binom{n}{(n+1)/2}$.
- In any row, binomial coefficients first strictly increase to the middle term and then strictly decrease.
- r^{th} diagonal represents r -figurate numbers. 1st diagonal represent natural numbers, 2nd diagonal represent triangular numbers and so on.
- If a prime p occurs n^{th} row in the 1st diagonal, it divides all other elements in n^{th} row except the 1's at ends.
- All the entries in r^{th} -diagonal of the left-justified Pascal triangle sum to r^{th} Fibonacci number.
- The rule $\sum_{k=r}^n \binom{k}{r} = \binom{n+1}{r+1}$ can be nicely identified on the Pascal triangle as shown. Due to its shape, it is known as Hockey-stick rule.



Applications

1. Given below is a row of Pascal's Triangle. Fill in the remaining numbers.

1 ——— ——— ——— 126 ——— 84 ——— ——— 1

2. The following is a portion of Pascal's triangle. Find the values of X, Y and Z.

1287 ——— ——— X

$$\begin{array}{ccc} 3003 & 3432 & Y \\ \hline & 6435 & \\ & Z & \end{array}$$

- Argue that the sum of elements in any row of Pascal's triangle is twice the sum of elements in the previous row.
- Show that if p is a prime number occurring in any row, it divides all other entries of that row (except 1's at ends).
- Expand $(1 - 2x)^4$ using Pascal's triangle.
- Can you compute 11^6 using Pascal's triangle? How?
- Determine the number of shortest routes from the 1 at the top vertex of Pascal's triangle to 3rd element in 7th row, moving only along the elements of Pascal's triangle.

Multinomial Theorem

For all $n \geq 0$, $(x_1 + x_2 + \dots + x_r)^n = \sum_{a+b+\dots+r=n} \binom{n}{a,b,\dots,r} x_1^a x_2^b \dots x_r^r$

where $\binom{n}{a,b,\dots,r} = \frac{n!}{a!b!c!\dots r!}$.

- The coefficient of any term, $x_1^a x_2^b \dots x_r^r$ is $\binom{n}{a,b,\dots,r} = \frac{n!}{a!b!c!\dots r!}$.
- The total number of terms of a multinomial is given by ${}^{n+r-1}C_{r-1}$ or ${}^{n+r-1}C_n$.
- All the terms whose powers are permutations of each other will have same coefficient. Since, $\binom{n}{a,b,\dots,r} = \binom{n}{\text{Permutation}(a,b,\dots,r)}$

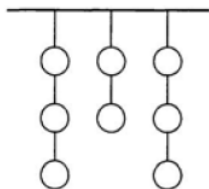
Applications

- Find the coefficient of $x^2 y^3$ in the expansion of
 - $(x + y + z)^4$
 - $(x + y + z)^5$
 - $(2x + y - 3)^6$
- How many terms are there in the expansion of $(a + b + c + d)^8$?
- Expand using multinomial theorem : $(x + y + z)^3$.
- $(1+x+x^2)^{20} = a_0 + a_1 x + a_2 x^2 + \dots + a_{39} x^{39} + a_{40} x^{40}$. Compute $(a_0 + a_1 + \dots + a_{40})$ and $(a_1 + a_3 + a_5 + \dots + a_{39})$.
- Show that the coefficients of each of ab^4 , ac^4 , bc^4 in the expansion $(a + b + c)^5$ are equal. Which other terms will have same coefficient as these?
- Of the 66 terms in the multinomial expansion of $(x + y + z)^{10}$, how many involve
 - just one variable?
 - exactly two variables?
 - all three variables?

Distribution into Groups

Applications

1. Suppose you have four boxes, labeled A, B, C, and D. How many ways are there to distribute (a) ten identical marbles among the four boxes? (b) the numbers 0–9 among the four boxes?
2. In how many ways can a group of six boys and six girls be divided into two equal groups so that each group contains odd numbers of boys and girls?
3. How many 27-digit natural numbers are there, containing the digits 1, 2, ..., 9 if each digit appears three times?
4. In a shooting tournament, eight clay targets are arranged in three hanging columns, as shown. A marksman is to break all eight targets according to the following rule : “One must break the lowest remaining target before breaking higher unbroken targets in the same column”. He is free to switch columns for the next shot. In how many different ways, can these targets be broken?



5. There are fifteen boys attending the Soccer Camp. Each of them is assigned to work with one of the five coaches. It is noted that each coach works with a different number of boys. In how many ways can the groups of boys be assigned?
6. In how many ways can 4 double-scoop ice-creams be ordered from a menu of 12 different flavors when the order of scoops is not taken into consideration?
7. Show combinatorially that $(4!)^{3!}$ is a divisor of $(4!)!$.
8. In how many ways can 7 balls be distributed in 3 different urns if
 - (a) all balls are different and any urn can contain any number of balls?
 - (b) the balls are indistinguishable and any urn can contain any number of balls?
 - (c) the balls are indistinguishable and any urn can contain no more than one ball?

Bijection Principle

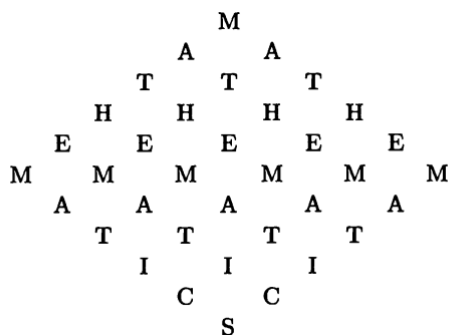
A relation between two sets is known as a bijection when it is both one-one and onto. Since each distinct element of set 1 is related to a distinct element of set 2 (one-one correspondence) and the relation has been defined for all of the elements (onto relation), clearly, the number of elements in both the subsets must be the same.

The “Bijection Principle” states that if a bijection can be established between two sets A and B , then their cardinalities are the same.

$$|A| = |B|$$

Applications

1. Show that there exists a bijection between subsets of an n -element set and n -digit binary sequences. Thus find out the total number of subsets of an n -element set.
2. A medical student has to work at a hospital for five days in January. However he is not allowed to work for two consecutive days. In how many different ways can he choose to work at the hospital?
3. In the following figure, in how many ways can one spell “MATHEMATICS” while traversing one con-



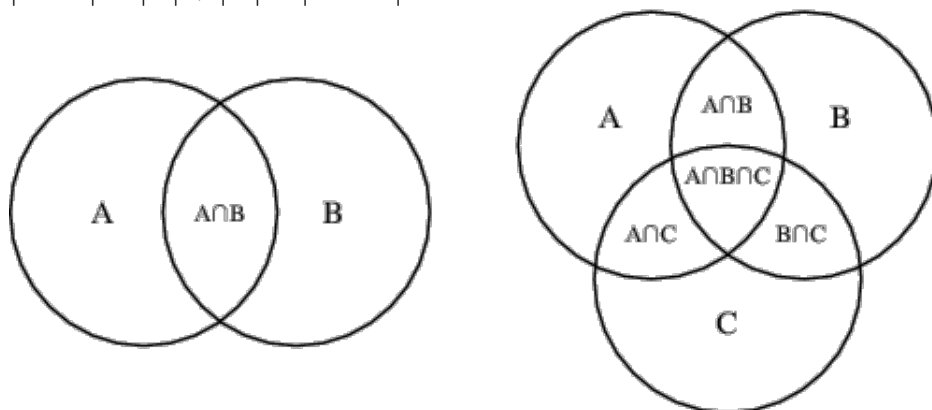
tinuous path?

4. Let X be a set containing n elements. Find the number of all ordered triples (A, B, C) of subsets of X such that A is a subset of B and B is a proper subset of C .
5. Let n be a positive integer. In how many ways can one write a sum of (atleast two) positive integers that add upto n ? Consider the same set of integers written in a different order as being different. (For example, there are 3 ways to express 3 as $3 = 1 + 1 + 1 = 1 + 2 = 2 + 1$)
6. A is $\{1, 2, 3, \dots, 100\}$. Find the number of 20-element subsets B of A such that the difference between any two elements of B is atleast two.

The Principle of Inclusion and Exclusion

The Principle of Inclusion and Exclusion (PIE): Let S be a finite set of objects. Let A and B be two subsets of S . Then to count the number of elements in $A \cup B$ is to count the number of elements of A and those of B , and subtract the number of elements common to both A and B , since they have been counted twice.

$$|A \cup B| = |A| + |B| - |A \cap B|$$



For three subsets, A , B and C of S , similarly

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |B \cap C| - |C \cap A| + |A \cap B \cap C|$$

In general,

$$|A_1 \cup A_2 \cup \cdots \cup A_n| = \sum |A_i| - \sum |A_i \cap A_j| + \sum |A_i \cap A_j \cap A_k| - \cdots + (-1)^{n-1} \sum |A_1 \cap A_2 \cap \cdots \cap A_n|$$

Applications

1. How many natural numbers less than 201 are divisible by none of 2 or 3 or 5?
2. Find the number of rearrangements of the word ABCDEFG that contain each of the following. (a) The sequence ABC. (b) The sequences AB, CD, and EF. (c) The sequences AB, BC, and EF.
3. How many 5-digit numbers do not contain the subsequence 121?
4. Find the number of rearrangements of the string 12345 in which none of the sequences 12, 23 and 34 occur.
5. Let $A = \{1,2,3,4,5,6,7,8\}$. Find the number of subsets of A which are neither subsets of $\{1,2,3,4,5\}$ nor of $\{4,5,6,7,8\}$.
6. Find the number of five-letter words that use letters from the set $\{A, B, C\}$ and include at least one A and at least one B.
7. Consider a sequence of natural numbers which is obtained by deleting all multiples of 3 or 4 or 5 less than 610. Which term of this sequence is 601?
8. Find the number of rearrangements of 12345 such that 1 is not in position 3, 2 is not in position 1 and 4 is not in position 4.
9. A faulty car odometer proceeds from digit 3 to digit 5, always skipping the digit 4, regardless of position. If the odometer now reads 002005, how many miles has the car actually traveled?

Derangements

Consider a permutation “2314” of the elements of $\{1,2,3,4\}$. Here 2 occurs at first position, 3 occurs at second, 1 at third, and 4 at fourth position. Then all the permutations in which 2 does not occur at first position, 3 does not occur at second, 1 not at third and 4 not at fourth, are known as derangements of the original permutation (here “2314”). For example, 3421 and 4132 are derangements of 2314, while 3412 and 1234 are not.

A “*derangement*” of a particular permutation is a set of all the permutations in which none of the elements occupy their original places.

Applications

1. List all derangements of ABCD.
2. A postman has to deliver four couriers to four registered addresses. In how many ways can he deliver such that
 - (a) all 4
 - (b) only 3
 - (c) only 2
 - (d) only 1

(e) none

of the couriers get delivered to correct addresses?

3. Six people arrive at a party with their own umbrellas. After the party, in how many ways can they leave such that two persons carry their own umbrellas while none of the remaining leaves with his own umbrella?
4. Find the number of permutations of the string 1234 in which at least one of the digits is in its original position.
5. In our village we have a baker, a grocer, a publican, and a policeman. They are called Mr. Bun, Mr. Sugar, Mr. Pale-Ale, and Mr. Copper, but Mr. Bun is not the baker, Mr. Sugar is not the grocer, Mr. Pale-Ale is not the publican, and Mr. Copper is not the policeman. How many possibilities remain for the last names of the baker, grocer, publican, and policeman?
6. On the way into a party 5 people check a coat and a bag at the door each. On the way out, the attendant hands out coats and bags randomly. In how many ways can this be done if no one gets either their own coat or their own bag?
7. Eight people are sitting in a row. In how many ways one can switch the seats of four of them, while the remaining four stay in their seats?

Pigeon Hole Principle

The simplest version of Dirichlet's box principle reads as follows:

- If $(n + 1)$ pearls are put into n boxes, then at least one box has more than one pearl.

The generalized Dirichlet's box principle or Pigeonhole Principle as it is commonly known now is as follows.

- If $kn + 1$ objects ($k \geq 1$ not necessarily finite) are distributed among n boxes, one of the boxes will contain at least $k + 1$ objects.
- This can also be stated as, if k discrete objects are to be allocated to n containers, then at least one container must hold at least $\lceil k/n \rceil$ objects, where $\lceil x \rceil$ is the ceiling function, denoting the smallest integer larger than or equal to x . Similarly, at least one container must hold no more than $\lfloor k/n \rfloor$ objects, where $\lfloor x \rfloor$ is the floor function, denoting the largest integer smaller than or equal to x .

Many problems can be solved by the application of the following forms of PHP.

- If n numbers have sum S , then, at least one of them must be at least $\frac{S}{n}$.
- If n numbers have product P , then, at least one of them must be at least $\sqrt[n]{P}$.
- Strong form: Let $q_1, q_2, \dots, q_n$ be positive integers. If $q_1 + q_2 + \dots + q_n - n + 1$ objects are distributed into n boxes, then either the first box contains at least q_1 objects, or the second box contains at least q_2 objects, ..., or the n^{th} box contains at least q_n objects.

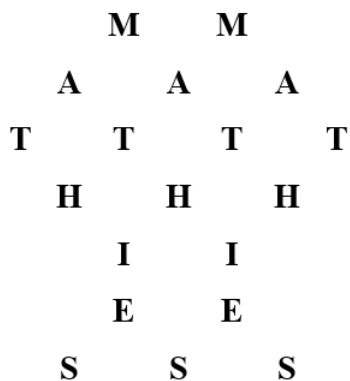
Applications

1. Show that in a group of n people there will be two who have equal number of friends.
2. Prove that there exists two powers of 2 that differ by a multiple of 2017.
3. Fifteen boys gathered 100 nuts. Prove that some pair of boys gathered identical number of nuts.
4. S is a set of n positive integers. None of the elements of S is divisible by n . Prove that there exists a subset of S such that the sum of its elements is divisible by n .
5. Five *lattice* points are chosen on an infinite square lattice. Prove that the midpoint of one of the segment joining two of these points is also lattice point.
6. A busy airport sees 1500 takeoffs per day. Prove that there are two planes that must take off within a minute of each other.
7. The set M consists of nine positive integers, none of which has a prime divisor larger than six. Prove that M has two elements whose product is the square of an integer.
8. Prove that one of $(n + 1)$ numbers from $\{1, 2, \dots, 2n\}$ is divisible by another.
9. Inside a room of area 5, you place 9 rugs, each of area 1 and an arbitrary shape. Prove that there are two rugs which overlap by at least $1/9$.

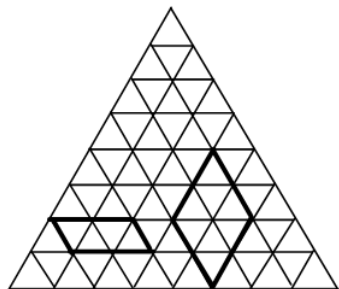
Exercises

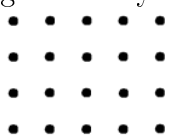
1. A spider has one sock and one shoe for each of its eight legs. In how many different orders can the spider put on its socks and shoes, assuming that, on each leg, the sock must be put on before the shoe?
2. In how many ways can the following pieces (one white and one black) be placed on a standard chessboard so they do not attack each other?
 - (a) two bishops
 - (b) two queens
 - (c) two knights
3. Determine the number of functions $f : \{1, 2, \dots, 1999\} \rightarrow \{2000, 2001, 2002, 2003\}$ satisfying the condition that $f(1) + f(2) + \dots + f(1999)$ is odd.
4. Define a *good word* as a sequence of letters that consists only of the letters A , B , and C - some of these letters may not appear in the sequence - and in which A is never immediately followed by B , B is never immediately followed by C , and C is never immediately followed by A . How many seven-letter good words are there?

5. In how many ways can one spell "MATHIES" in the following diagram by traversing along a continuous line?

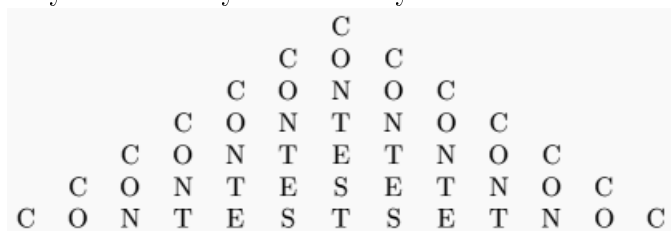


6. A triangular grid is obtained by tiling an equilateral triangle of side length 8 by equilateral triangles of side length 1. Determine the number of parallelograms bounded by line segments of the grid.



7. A 4×5 rectangular array of points, each of which is 1 unit away from its nearest neighbors. Determine the number of non-degenerate triangles (ie, triangles with positive area) whose vertices are points in the given array.
- 
8. A positive integer is called *ascending* if, in its decimal representation, there are at least two digits and each digit is less than any digit to its right. How many ascending positive integers are there?
9. How many ordered triplets of positive integers are solutions of $abc = 210$?
10. A 10-digit positive integer is called a *cute* number if its digits are from the set $\{1, 2, 3\}$ and every two consecutive digits differ by 1. Find the total number of cute numbers. Prove that the sum of all 10-digit cute numbers is divisible by 1408.
11. Two n digit positive integers are said to be equivalent if one is a permutation of the other (leading zeroes allowed). For example, 10075, 01057, 00751 are equivalent. Find the number of 5 digit positive numbers of which, no two are equivalent.
12. Given a rational number, write it as a fraction in lowest terms and calculate the product of the resulting numerator and denominator. For how many rational numbers between 0 and 1 will $20!$ be the resulting product?
13. How many isosceles triangles are there, such that each of its sides are of integer length, and its perimeter is 100?

14. Find the number of acute angled triangles whose sides are three consecutive integers and perimeter is not more than 100.
15. For how many pairs of consecutive integers in $\{1000, 1001, 1002, \dots, 2000\}$ is no carrying required when the two integers are added?
16. Compute the number of sets of three distinct elements that can be chosen from the set $\{2^1, 2^2, 2^3, \dots, 2^{200}\}$ such that the three elements form a geometric progression.
17. Find the number of ways to tile a 6×2 rectangle with dominos. (A domino is a 2×1 or 1×2 rectangle, assume all dominos to be identical).
18. How many subsets of the set $\{1, 2, \dots, 10\}$ contain at least one odd integer?
19. In Mr Fat's math class, there are five boys and nine girls. At the end of the term, Mr Fat wants to take a picture of the whole class. He wants all students to stand in a row, with boys standing in a decreasing order from left to right and girls standing in increasing order according to their height (assume all students have distinct heights). In how many ways can this be done? (The boys need not stand together, and the girls need not stand together.)
20. For a game, nine tiles are numbered $1, 2, 3, \dots, 9$. Each of three players randomly selects and keeps three of the tiles and sums the three values. In how many ways can all three players obtain odd sums?
21. Let n be a 5-digit number, and let q and r be the quotient and remainder, respectively, when n is divided by 100. For how many values of n is $q + r$ divisible by 4?
22. How many ways are there to arrange 5 A's, 5 B's and 5 C's in a row such that there is no A in first five letters, no B in middle five letters and no C in last five letters?
23. An integer between 1000 and 9999, inclusive, is called *balanced* if the sum of its two leftmost digits equals the sum of its two rightmost digits. How many balanced integers are there?
24. Find the number of ways the words "CONTEST" can be spelled in the following diagram, while moving only horizontally or vertically.



25. In how many ways may 1, 2, 3, 4, 5 and 6 be ordered so that no two consecutive elements have a sum that is divisible by 2 or 3?
26. What is the minimum of number of lattice points to be taken in an infinite cubic lattice to ensure that the mid-point of any one of the line-segments formed by joining these points is also a lattice point?
27. In the sequence 1, 1, 2, 3, 5, 8, 3, 1, 4, ... each term starting with the third is the sum of the two preceding terms. But addition is done mod 10. Prove that the sequence is purely periodic. What is the maximum possible length of the period?
28. Provide a combinatorial proof for Fermat's Little Theorem which states $a^p - a$ is divisible by a prime p for any natural number a .

Window to RMO

PreRMO

1. (preRMO 2012) For how many pairs of positive integers (x, y) is $x + 3y = 100$?
2. (preRMO 2012) A postman has to deliver five letters to five different houses. Mischievously, he posts one letter through each door without looking to see if it is the correct address. In how many different ways could he do this so that exactly two of the five houses receive the correct letters?
3. (preRMO 2013) There are $n - 1$ red balls, n green balls and $n + 1$ blue balls in a bag. The number of ways of choosing two balls from the bag that have different colours is 299. What is the value of n ?
4. (preRMO 2013) To each element of the set $S = \{1, 2, \dots, 1000\}$ a colour is assigned. Suppose that for any two elements a, b of S , if 15 divides $a + b$ then they are both assigned the same colour. What is the maximum possible number of distinct colours used?
5. (preRMO 2013) What is the sum (in base 10) of all the natural numbers less than 64 which have exactly three ones in their base 2 representation?
6. (preRMO 2014) What is the number of ordered pairs (A, B) where A and B are subsets of $\{1, 2, \dots, 5\}$ such that neither $A \subseteq B$ nor $B \subseteq A$?
7. (preRMO 2015) How many line segments have both their endpoints located at the vertices of a given cube?
8. (preRMO 2015) A subset B of the set of first 100 positive integers has the property that no two elements of B sum to 125. What is the maximum possible number of elements in B ?

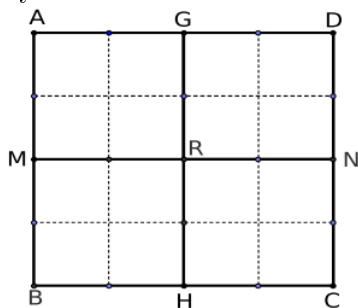
RMO

1. (RMO 1990) Two boxes contain between them 65 balls of several different sizes. Each ball is white, black, red or yellow. If you take any five balls of the same colour, at least two of them will always be of the same size(radius). Prove that there are at least three ball which lie in the same box have the same colour and have the same size(radius).
2. (RMO 1991) There are two urns each containing an arbitrary number of balls. We are allowed two types of operations: (a) remove an equal number of balls simultaneously from both the urns, (b) double the number of balls in any of them Show that after performing these operations finitely many times, both the urns can be made empty.
3. (RMO 1991) The 64 squares of an 8×8 chessboard are filled with positive integers in such a way that each integer is the average of the integers on the neighboring squares. Show that in fact all the 64 entries are equal.
4. (RMO 1993) Suppose $A_1, A_2, A_3, \dots, A_{20}$ is a 20 sides regular polygon. How many non-isosceles (scalene) triangles can be formed whose vertices are among the vertices of the polygon but the sides are not the sides of the polygon?
5. (RMO 1994) Let A be a set of 16 positive integers with the property that the product of any two distinct members of A will not exceed 1994. Show that there are numbers a and b in the set A such that the gcd of a and b is greater than 1.

6. (RMO 1994) Find all rational numbers $\frac{m}{n}$ such that
 - (i) $0 < \frac{m}{n} < 1$;
 - (ii) m and n are relatively prime;
 - (iii) $mn = 25!$.
7. (RMO 1995) Let $A_1 A_2 A_3 \dots A_{21}$ be a 21-sided regular polygon inscribed in a circle with centre O . How many triangles $A_i A_j A_k$, $1 \leq i < j < k \leq 21$, contain the centre point O in their interior?
8. (RMO 1996) If A is a fifty element subset of the set $1, 2, \dots, 100$ such that no two numbers from A add up to 100, show that A contains a square.
9. (RMO 1997) Find the number of unordered pairs $\{A, B\}$ of subsets of an n -element set X that satisfies the following: (a) $A \neq B$ (b) $A \cup B = X$.
10. (RMO 1998) Given the 7-element set $A = \{a, b, c, d, e, f, g\}$, find a collection T of 3-element subsets of A such that each pair of elements from A occurs exactly once on one of the subsets of T .
11. (RMO 1999) Find the number of quadratic polynomials $ax^2 + bx + c$ which satisfy the following:
 - (a) a, b, c are distinct;
 - (b) $a, b, c \in \{1, 2, 3, \dots, 1999\}$;
 - (c) $x + 1$ divides $ax^2 + bx + c$.
12. (RMO 2000) All the 7 digit numbers containing each of the digits 1, 2, 3, 4, 5, 6, 7 exactly once, and not divisible by 5 are arranged in increasing order. Find the 200th number in the list.
13. (RMO 2001) Consider an $n \times n$ array of numbers a_{ij} (standard notation). Suppose each row consists of the n numbers $1, 2, \dots, n$ in some order and $a_{ij} = a_{ji}$ for $i, j = 1, 2, \dots, n$. If n is odd, prove that the numbers $a_{11}, a_{22}, \dots, a_{nn}$ are $1, 2, 3, \dots, n$ in some order.
14. (RMO 2002) Suppose the integers $1, 2, \dots, 10$ are split into two disjoint collections $a_1, a_2, \dots, a_5$ and $b_1, \dots, b_5$ such that $a_1 < a_2 < a_3 < a_4 < a_5, b_1 > b_2 > b_3 > b_4 > b_5$
 - (i) Show that the larger number in any pair $\{a_j, b_j\}$, $1 \leq j \leq 5$ is at least 6.
 - (ii) Show that $\sum_{i=1}^5 |a_i - b_i| = 25$ for every such partition.
15. (RMO 2003) Consider the set $X = \{1, 2, \dots, 10\}$. Find two disjoint nonempty subsets A and B of X such that
 - a) $A \cup B = X$;
 - b) $\prod_{x \in A} x$ is divisible by $\prod_{x \in B} x$, where $\prod_{x \in C} x$ is the product of all numbers in C ;
 - c) $\frac{\prod_{x \in A} x}{\prod_{x \in B} x}$ is as small as possible.
16. (RMO 2003) Find the number of ordered triples (x, y, z) of non-negative integers satisfying
 - (i) $x \leq y \leq z$ (ii) $x + y + z \leq 100$.
17. (RMO 2004) Prove that the number of triples (A, B, C) where A, B, C are subsets of $\{1, 2, \dots, n\}$ such that $A \cap B \cap C = \phi$, $A \cap B \neq \phi$, $C \cap B \neq \phi$ is $7^n - 2 \cdot 6^n + 5^n$.
18. (RMO 2005) Find the number of 5-digit numbers that each contains the block '15' and is divisible by 15.
19. (RMO 2006) A 6×6 square is dissected into 9 rectangles by lines parallel to its sides such that all these rectangles have integer sides. Prove that there are always two congruent rectangles.

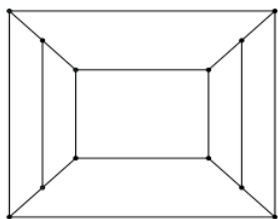
20. (RMO 2007) How many 6 – *digit* numbers are there such that-: a)The digits of each number are all from the set $\{1, 2, 3, 4, 5\}$ b)any digit that appears in the number appears at least twice ? (Example: 225252 is valid while 222133 is not)
21. (RMO 2008) Find the number of all 6-digit natural numbers such that the sum of their digits is 10 and each of the digits 0, 1, 2, 3 occurs at least once in them.
22. (RMO 2008) Find the number of all integer-sided isosceles obtuse-angled triangles with perimeter 2008.
23. (RMO 2009) Find the sum of all 3-digit natural numbers which contain at least one odd digit and at least one even digit.
24. (RMO 2010) Find the number of 4-digit numbers (in base 10) having non-zero digits and which are divisible by 4 but not by 8.
25. (RMO 2011 paper1) Consider a 20-sided convex polygon K , with vertices $A_1, A_2, \dots, A_{20}$ in that order. Find the number of ways in which three sides of K can be chosen so that every pair among them has at least two sides of K between them. (For example $(A_1A_2, A_4A_5, A_{11}A_{12})$ is an admissible triple while $(A_1A_2, A_4A_5, A_{19}A_{20})$ is not.)
26. Let $(a_1, a_2, a_3, \dots, a_{2011})$ be a permutation of the numbers $1, 2, 3, \dots, 2011$. Show that there exist two numbers j, k such that $1 \leq j < k \leq 2011$ and $|a_j - j| = |a_k - k|$
27. (RMO 2012 paper 1) Let $X = 1, 2, 3, \dots, 10$. Find the the number of pairs $\{A, B\}$ such that $A \subseteq X, B \subseteq X, A \neq B$ and $A \cap B = \{2, 3, 5, 7\}$.
28. (RMO 2012 paper 2) Let $X = \{1, 2, 3, \dots, 12\}$. Find the the number of pairs $\{A, B\}$ such that $A \subseteq X, B \subseteq X, A \neq B$ and $A \cap B = \{2, 3, 5, 7, 8\}$.
29. (RMO 2012 paper 3) Let $X = \{1, 2, 3, \dots, 10\}$. Find the the number of pairs $\{A, B\}$ such that $A \subseteq X, B \subseteq X, A \neq B$ and $A \cap B = \{5, 7, 8\}$.
30. (RMO 2012 paper 4) Let $X = \{1, 2, 3, \dots, 11\}$. Find the the number of pairs $\{A, B\}$ such that $A \subseteq X, B \subseteq X, A \neq B$ and $A \cap B = \{4, 5, 7, 8, 9, 10\}$.
31. (RMO 2012 Mumbai region) Let $X = \{1, 2, 3, \dots, 11\}$. Find the the number of pairs $\{A, B\}$ such that $A \subseteq X, B \subseteq X, A \neq B$ and $A \cap B = \{4, 5, 7, 8, 9, 10\}$.
32. (RMO 2012 regional) A computer program generated 175 positive integers at random, none of which had a prime divisor greater than 10. Prove that there are three numbers among them whose product is the cube of an integer.
33. (RMO 2013 paper1) Find the number of 10-tuples $(a_1, a_2, \dots, a_{10})$ of integers such that $|a_i| \leq 1$ and $a_1^2 + a_2^2 + a_3^2 + \dots + a_{10}^2 - a_1a_2 - a_2a_3 - a_3a_4 - \dots - a_9a_{10} - a_{10}a_1 = 2$.
34. (RMO 2013 paper 2) For a natural number n , let $T(n)$ denote the number of ways we can place n objects of weights $1, 2, \dots, n$ on a balance such that the sum of the weights in each pan is the same. Prove that $T(100) > T(99)$.
35. (RMO 2013 paper 3) Find the number of eight-digit numbers the sum of whose digits is 4
36. (RMO 2013 paper 4) Suppose that the vertices of a regular polygon of 20 sides are coloured with three colours – red, blue and green – such that there are exactly three red vertices. Prove that there are three vertices A,B,C of the polygon having the same colour such that triangle ABC is isosceles.

37. (RMO 2013 Mumbai Region) A finite non-empty set S of integers is called 3-good if the sum of the elements of S is divisible by 3. Find the number of 3-good non-empty subsets of $\{0, 1, 2, \dots, 9\}$.
38. (RMO 2013 Regional) Let $n \geq 4$ be a natural number. Let $A_1 A_2 \dots A_n$ be a regular polygon and $X = \{1, 2, 3, \dots, n\}$. A subset $\{i_1, i_2, \dots, i_k\}$ of X , with $k \geq 3$ and $i_1 < i_2 < \dots < i_k$, is called a good subset if the angles of the polygon $A_{i_1} A_{i_2} \dots A_{i_k}$, when arranged in the increasing order, are in an arithmetic progression. If n is a prime, show that a proper good subset of X contains exactly four elements.
39. (RMO 2014 paper1) Is it possible to write the numbers $17, 18, 19, \dots, 32$ in a 4×4 grid of unit squares, with one number in each square, such that the product of the numbers in each 2×2 sub-grids is divisible by 16?



40. (RMO 2014 paper2) Suppose n is odd and each square of an $n \times n$ grid is arbitrarily filled with either by 1 or by -1 . Let r_j and c_k denote the product of all numbers in j -th row and k -th column respectively, $1 \leq j, k \leq n$. Prove that
- $$\sum_{j=1}^n r_j + \sum_{k=1}^n c_k \neq 0$$
41. (RMO 2014 paper3) For any natural number n , let $S(n)$ denote the sum of the digits of n . Find the number of all 3-digit numbers n such that $S(S(n)) = 2$.

42. (RMO 2014 paper4) In the adjacent figure, can the numbers $1, 2, 3, 4, \dots, 18$ be placed, one on each line segment, such that the sum of the numbers on the three line segments meeting at each point is divisible by 3?



43. (RMO 2014 MUMBAI REGION) A person moves in the $x - y$ plane moving along points with integer co-ordinates x and y only. When she is at point (x, y) , she takes a step based on the following rules:
- (a) if $x + y$ is even she moves to either $(x + 1, y)$ or $(x + 1, y + 1)$;
 - (b) if $x + y$ is odd she moves to either $(x, y + 1)$ or $(x + 1, y + 1)$.
- How many distinct paths can she take to go from $(0, 0)$ to $(8, 8)$ given that she took exactly three steps to the right ((x, y) to $(x + 1, y)$)?
44. (RMO 2015 Ahmadnagar) Suppose 40 objects are placed among a circle at equal distance. In how many ways can 3 objects be chosen from among them so that no two of the three objects are adjacent nor diametrically opposite?

45. (RMO 2015 paper 1) Suppose 28 objects are placed along a circle at equal distances. In how many ways can 3 objects be chosen from among them so that no two of the three chosen objects are adjacent nor diametrically opposite?
46. (RMO 2015 paper 2) Suppose 32 objects are placed along a circle at equal distances. In how many ways can 3 objects be chosen from among them so that no two of the three chosen objects are adjacent nor diametrically opposite?
47. (RMO 2015 paper 3) Suppose 36 objects are placed along a circle at equal distances. In how many ways can 3 objects be chosen from among them so that no two of the three chosen objects are adjacent nor diametrically opposite?
48. (RMO 2015 paper 4) Suppose 40 objects are placed along a circle at equal distances. In how many ways can 3 objects be chosen from among them so that no two of the three chosen objects are adjacent nor diametrically opposite?
49. (RMO mumbai 2016) Find all 6 digit natural numbers such that 3 of its digits are odd and rest 3 are even.
50. (RMO mumbai 2016) Let $S = 1, 2, \dots, n$ and let T be the set of all ordered triples of subsets of S , say (A_1, A_2, A_3) , such that $A_1 \cup A_2 \cup A_3 = S$. Determine, in terms of n ,

$$\sum_{(A_1, A_2, A_3) \in T} |A_1 \cap A_2 \cap A_3|$$

where $|X|$ denotes the number of elements in the set X . (For example, if $S = 1, 2, 3$ and $A_1 = 1, 2$, $A_2 = 2, 3$, $A_3 = 3$ then one of the elements of T is $(1, 2, 2, 3, 3)$.)

51. (RMO 2016) At an international event there are 100 countries participating, each with its own flag. There are 10 distinct flagpoles at the stadium, labelled #1, #2, ..., #10 in a row. In how many ways can all the 100 flags be hoisted on these 10 flagpoles, such that for each i from 1 to 10, the flagpole # i has at least i flags? (Note that the vertical order of the flagpoles on each flag is important)
52. (RMO 2016 Delhi Region) A deck of 52 cards is given. There are four suites each having cards numbered $1, 2, \dots, 13$. The audience chooses some five cards with distinct numbers written on them. The assistant of the magician comes by, looks at the five cards and turns exactly one of them face down and arranges all five cards in some order. Then the magician enters and with an agreement made beforehand with the assistant, he has to determine the face down card (both suite and number). Explain how the trick can be completed.
53. (RMO 2016 Delhi Region) a.) A 7-tuple $(a_1, a_2, a_3, a_4, b_1, b_2, b_3)$ of pairwise distinct positive integers with no common factor is called a shy tuple if

$$a_1^2 + a_2^2 + a_3^2 + a_4^2 = b_1^2 + b_2^2 + b_3^2$$

and for all $1 \leq i < j \leq 4$ and $1 \leq k \leq 3$, $a_i^2 + a_j^2 \neq b_k^2$. Prove that there exists infinitely many shy tuples.

b) Show that 2016 can be written as a sum of squares of four distinct natural numbers.

Past RMO Papers

RMO 1990

1. Two boxes contain between them 65 balls of several different sizes. Each ball is white, black, red or yellow. If you take any five balls of the same colour, at least two of them will always be of the same size(radius). Prove that there are at least three ball which lie in the same box have the same colour and have the same size(radius).
2. For all positive real numbers a, b, c , prove that

$$\frac{a}{b+c} + \frac{b}{c+a} + \frac{c}{a+b} \geq \frac{3}{2}.$$

3. A square sheet of paper $ABCD$ is so folded that B falls on the mid point of M of CD . Prove that the crease will divide BC in the ratio 5 : 3.
4. Find the remainder when 2^{1990} is divided by 1990.
5. P is any point inside a triangle ABC . The perimeter of the triangle $AB + BC + Ca = 2s$. Prove that $s < AP + BP + CP < 2s$.
6. N is a 50-digit number (in decimal representation). All digits except the 26th digit (from the left) are 1. If N is divisible by 13, find its 26-th digit.
7. A census man on duty visited a house in which the lady inmates declined to reveal their individual ages, but said "We do not mind giving you the sum of the ages of any two ladies you may choose". Thereupon, the census man said, "In that case, please give me the sum of the ages of every possible pair of you". They gave the sums as: 30, 33, 41, 58, 66, 69. The census man took these figures and happily went away. How did he calculate the individual ages?
8. If the circumcenter and centroid of a triangle coincide, prove that it must be equilateral.

RMO 1991

1. Let P be an interior point of a triangle ABC and AP, BP, CP meet the sides BC, CA, AB in D, E, F respectively. Show that

$$\frac{AP}{PD} = \frac{AF}{FB} + \frac{AE}{EC}.$$

2. If a, b, c, d be any four positive real numbers, then prove that

$$\frac{a}{b} + \frac{b}{c} + \frac{c}{d} + \frac{d}{a} \geq 4.$$

3. A four-digit number has the following properties:

- (a) It is a perfect square;
- (b) Its first two digits are equal
- (c) Its last two digits are equal.

Find all such four-digit numbers.

4. There are two urns each containing an arbitrary number of balls. We are allowed two types of operations: (a) remove an equal number of balls simultaneously from both the urns, (b) double the number of balls in any of them. Show that after performing these operations finitely many times, both the urns can be made empty.
5. Take any point P_1 on the side BC of a triangle ABC and draw the following chain of lines: P_1P_2 parallel to AC ; P_2P_3 parallel to BC ; P_3P_4 parallel to AB ; P_4P_5 parallel to CA ; and P_5P_6 parallel to BC . Here, P_2, P_5 lie on AB ; P_3, P_6 lie on CA and P_4 on BC . Show that P_6P_1 is parallel to AB .
6. Find all integer values of a such that the quadratic expression $(x + a)(x + 1991) + 1$ can be factored as a product $(x + b)(x + c)$ where b, c are integers.
7. Prove that $n^4 + 4^n$ is composite for all values of n greater than 1.
8. The 64 squares of an 8×8 chessboard are filled with positive integers in such a way that each integer is the average of the integers on the neighbouring squares. Show that in fact all the 64 entries are equal.

RMO 1992

1. Determine the set of integers n for which $n^2 + 19n + 92$ is a square.
2. If $\frac{1}{a} + \frac{1}{b} = \frac{1}{c}$, where a, b, c are positive integers with no common factor, prove that $(a + b)$ is a square.
3. Determine the largest 3 digit prime number that is a factor of $\binom{2000}{1000}$.
4. $ABCD$ is a cyclic quadrilateral with $AC \perp BD$; AC meets BD at E . Prove that

$$EA^2 + EB^2 + EC^2 + ED^2 = 4R^2$$

where R is the radius of the circumscribing circle.

5. $ABCD$ is a quadrilateral and P, Q are the midpoints of CD, AB , AP, DQ meet at X and BP, CQ meet at Y . Prove that $A[ADX] + A[BCY] = A[PXOY]$.
6. Prove that

$$1 < \frac{1}{1001} + \frac{1}{1002} + \frac{1}{1003} + \cdots + \frac{1}{3001} < 1\frac{1}{3}.$$

7. Solve the system

$$\begin{aligned}(x + y)(x + y + z) &= 18 \\ (y + z)(x + y + z) &= 30 \\ (x + z)(x + y + z) &= 2A\end{aligned}$$

in terms of the parameter A .

8. The cyclic octagon $ABCDEFGH$ has sides a, a, a, a, b, b, b, b respectively. Find the radius of the circle that circumscribes $ABCDEFGH$.

RMO 1993

1. Let ABC be an acute angled triangle and CD be the altitude through C . If $AB = 8$ and $CD = 6$, find the distance between the midpoints of AD and BC .
2. Prove that the ten's digit of any power of 3 is even.
3. Suppose $A_1, A_2, A_3, \dots, A_{20}$ is a 20 sides regular polygon. How many non-isosceles (scalene) triangles can be formed whose vertices are among the vertices of the polygon but the sides are not the sides of the polygon?
4. Let $ABCD$ be a rectangle with $AB = a$ and $BC = b$. Suppose r_1 is the radius of the circle passing through A and B and touching CD ; and similarly r_2 is the radius of the circle passing through B and C and touching AD . Show that

$$r_1 + r_2 \geq \frac{5}{8}(a + b).$$

5. Show that $19^{93} - 13^{99}$ is a positive integer divisible by 162.
6. If a, b, c, d are four positive reals such that $abcd = 1$, prove that $(1 + a)(1 + b)(1 + c)(1 + d) \geq 16$.
7. In the group of ten persons, each person is asked to write the sum of the ages of all the other nine persons. Of all ten sums form the nine-element set $\{82, 83, 84, 85, 87, 89, 90, 91, 92\}$, find the individual ages of the persons, assuming them to be whole numbers.

RMO 1994

1. A leaf is torn from a paperback novel. The sum of the numbers on the remaining pages is 15000. What are the page numbers on the torn leaf?
2. In a triangle ABC , the incircle touches the sides BC, CA, AB at D, E, F respectively. If the radius of the incircle is 4 units and if BD, CE, AF are consecutive integers, find the sides of the triangle ABC .
3. Find all 6-digit numbers $a_1a_2a_3a_4a_5a_6$ formed by using the digits 1, 2, 3, 4, 5, 6 once each such that the number $a_1a_2a_3 \dots a_k$ is divisible by k for $1 \leq k \leq 6$.
4. Solve the system of equations for real x and y :

$$\begin{aligned} 5x \left(1 + \frac{1}{x^2 + y^2} \right) &= 12 \\ 5y \left(1 - \frac{1}{x^2 + y^2} \right) &= 4. \end{aligned}$$

5. Let A be a set of 16 positive integers with the property that the product of any two distinct members of A will not exceed 1994. Show that there are numbers a and b in the set A such that the gcd of a and b is greater than 1.
6. Let AC and BD be two chords of a circle with center O such that they intersect at right angles inside the circle at the point M . Suppose K and L are midpoints of the chords AB and CD respectively. Prove that $OKML$ is a parallelogram.
7. Find all rational numbers $\frac{m}{n}$ such that
 - (a) $0 < \frac{m}{n} < 1$;
 - (b) m and n are relatively prime;
 - (c) $mn = 25!$.
8. If a, b, c are positive real numbers such that $a + b + c = 1$, prove that

$$(1 + a)(1 + b)(1 + c) \geq 8(1 - a)(1 - b)(1 - c).$$

RMO 1995

1. In triangle ABC , K and L are points on the side BC (K being closer to B than L) such that $BC \cdot KL = BK \cdot CL$ and AL bisects $\angle KAC$. Show that $AL \perp AB$.
2. Call a positive integer n good if there are n integers, positive or negative, and not necessarily distinct, such that their sum and products are both equal to n . Show that the integers of the form $4k + 1$ and $4l$ are good.
3. Prove that among any 18 consecutive three digit numbers there is at least one number which is divisible by the sum of its digits.
4. Show that the quadratic equation $x^2 + 7x - 14(q^2 + 1) = 0$, where q is an integer, has no integer root.
5. Show that for any triangle ABC , the following inequality is true:

$$a^2 + b^2 + c^2 > \sqrt{3} \max\{|a^2 - b^2|, |b^2 - c^2|, |c^2 - a^2|\}.$$

6. Let $A_1A_2A_3 \dots A_{21}$ be a 21-sided regular polygon inscribed in a circle with centre O . How many triangles $A_iA_jA_k$, $1 \leq i < j < k \leq 21$, contain the centre point O in their interior?
7. Show that for any real number x :

$$x^2 \sin x + x \cos x + x^2 + \frac{1}{2} > 0.$$

RMO 1996

1. The sides of a triangle are three consecutive integers and its inradius is 4. Find the circumradius.
2. Find all triples a, b, c of positive integers such that

$$\left(1 + \frac{1}{a}\right)\left(1 + \frac{1}{b}\right)\left(1 + \frac{1}{c}\right) = 3.$$

3. Solve for real numbers x and y ,

$$xy^2 = 15x^2 + 17xy + 15y^2;$$

$$x^2y = 20x^2 + 3y^2.$$

4. Suppose N is an n digit positive integer such that (a) all its digits are distinct; (b) the sum of any three consecutive digits is divisible by 5. Prove that $n \leq 6$. Further, show that starting with any digit, one can find a six digit number with these properties.
5. Let ABC be a triangle and h_a be the altitude through A . Prove that

$$(b + c)^2 \geq a^2 + 4h_a^2.$$

6. Given any positive integer n , show that there are two positive rational numbers a and b , $a \neq b$, which are not integers and which are such that $a - b, a^2 - b^2, \dots, a^n - b^n$ are all integers.
7. If A is a fifty element subset of the set $1, 2, \dots, 100$ such that no two numbers from A add up to 100, show that A contains a square.

RMO 1997

1. Let P be an interior point of a triangle ABC and let BP and CP meet AC and AB in E and F respectively. If $S_{BPF} = 4, S_{BPC} = 8$ and $S_{CPE} = 13$, find S_{AFPE} .
2. For each positive integer n , define $a_n = 20 + n^2$ and $d_n = \gcd(a_n, a_{n+1})$. Find the set of all values that are taken by d_n and show by examples that each of these values is attained.

3. Solve for real x :

$$\frac{1}{[x]} + \frac{1}{[2x]} = x - [x] + \frac{1}{3}.$$

4. In a quadrilateral $ABCD$, it is given that AB is parallel to CD and the diagonals AC and BD are perpendicular to each other. Show that (a) $AD \cdot BC \geq AB \cdot CD$ (b) $AD + BC \geq AB + CD$.
5. Let x, y, z be three distinct real positive numbers, Determine whether or not the three real numbers

$$\left| \frac{x}{y} - \frac{y}{x} \right|, \left| \frac{y}{z} - \frac{z}{y} \right|, \left| \frac{z}{x} - \frac{x}{z} \right|$$

can be the lengths of the sides of a triangle.

6. Find the number of unordered pairs $\{A, B\}$ of subsets of an n -element set X that satisfies the following:
(a) $A \neq B$ (b) $A \cup B = X$.

RMO 1998

1. Let $ABCD$ be a convex quadrilateral in which $\angle BAC = 50^\circ$, $\angle CAD = 60^\circ$ and $\angle BDC = 25^\circ$. If E is the point of intersection of AC and BD , find $\angle AEB$.
2. Let n be a positive integer and $p_1, p_2, p_3, \dots, p_n$ be n prime numbers all larger than 5 such that 6 divides $p_1^2 + p_2^2 + p_3^2 + \dots + p_n^2$. Prove that 6 divides n .
3. Prove that for every natural number $n > 1$

$$\frac{1}{n+1} \left(1 + \frac{1}{3} + \frac{1}{5} + \dots + \frac{1}{2n-1} \right) > \frac{1}{n} \left(\frac{1}{2} + \frac{1}{4} + \dots + \frac{1}{2n} \right).$$

4. Let ABC be a triangle with $AB = AC$ and $\angle BAC = 30^\circ$. Let A' be the reflection of A in the line BC ; B' be the reflection of B in the line CA ; C' be the reflection of C in line AB . Show that $A'B'C'$ is an equilateral triangle.
5. Find the minimum possible least common multiple of twenty natural numbers whose sum is 801.
6. Given the 7-element set $A = \{a, b, c, d, e, f, g\}$, find a collection T of 3-element subsets of A such that each pair of elements from A occurs exactly once on one of the subsets of T .

RMO 1999

1. Prove that the inradius of a right angled triangle with integer sides is an integer.
2. Find the number of positive integers which divide 10^{999} but not 10^{998} .
3. Let $ABCD$ be a square and M, N points on sides AB, BC respectively such that $\angle MDN = 45^\circ$. If R is the midpoint of MN show that $RP = RQ$ where P, Q are points of intersection of AC with the lines MD, ND .
4. If p, q, r are the roots of the cubic equation $x^3 - 3px^2 + 3q^2x - r^3 = 0$, then show that $p = q = r$.
5. If a, b, c are sides of a triangle, prove that

$$\frac{a}{c+a-b} + \frac{b}{a+b-c} + \frac{c}{b+c-a} \geq 3.$$

6. Find all solutions in integers m, n of the equation

$$(m-n)^2 = \frac{4mn}{m+n-1}.$$

7. Find the number of quadratic polynomials $ax^2 + bx + c$ which satisfy the following:
 - (a) a, b, c are distinct;
 - (b) $a, b, c \in \{1, 2, 3, \dots, 1999\}$;
 - (c) $x+1$ divides $ax^2 + bx + c$.

RMO 2000

1. Let AC be a line segment in the plane and B a points between A and C . Construct isosceles triangles PAB and QAC on one side of the segment AC such that $\angle APB = \angle BQC = 120^\circ$ and an isosceles triangle RAC on the other side of AC such that $\angle ARC = 120^\circ$. Show that PQR is an equilateral triangle.
2. Solve the equation $y^3 = x^3 + 8x^2 - 6x + 8$, for positive integers x and y .
3. Suppose $\{x_n\}_{n \geq 1}$ is a sequence of positive real numbers such that $x_1 \geq x_2 \geq x_3 \dots \geq x_n \dots$, and for all n

$$\frac{x_1}{1} + \frac{x_4}{2} + \frac{x_9}{3} + \dots + \frac{x_{n^2}}{n} \leq 1.$$

Show that for all k

$$\frac{x_1}{1} + \frac{x_2}{2} + \dots + \frac{x_k}{k} \leq 3.$$

4. All the 7 digit numbers containing each of the digits 1, 2, 3, 4, 5, 6, 7 exactly once , and not divisible by 5 are arranged in increasing order. Find the 200th number in the list.
5. The internal bisector of angle A in a triangle ABC with $AC > AB$ meets the circumcircle Γ of the triangle in D . Join D to the center O of the circle Γ and suppose that DO meets AC in E , possibly when extended. Given that BE is perpendicular to AD , show that AO is parallel to BD .
6. (i) Consider two positive integers a and b which are such that $a^a b^b$ is divisible by 2000. What is the least possible value of ab ? (ii) Consider two positive integers a and b which are such that $a^b b^a$ is divisible by 2000. What is the least possible value of ab ?
7. Find all real values of a such that $x^4 - 2ax^2 + x + a^2 - a = 0$ has all its roots real.

RMO 2001

1. Let BE and CF be the altitudes of an acute triangle ABC with E on AC and F on AB . Let O be the point of intersection of BE and CF . Take any line KL through O with K on AB and L on AC . Suppose M and N are located on BE and CF respectively, such that KM is perpendicular to BE and LN is perpendicular to CF . Prove that FM is parallel to EN .
2. Find all primes p and q such that $p^2 + 7pq + q^2$ is a perfect square.
3. Find the number of positive integers x such that

$$\left[\frac{x}{99} \right] = \left[\frac{x}{101} \right].$$

4. Consider an $n \times n$ array of numbers a_{ij} (standard notation). Suppose each row consists of the n numbers $1, 2, \dots, n$ in some order and $a_{ij} = a_{ji}$ for $i, j = 1, 2, \dots, n$. If n is odd, prove that the numbers $a_{11}, a_{22}, \dots, a_{nn}$ are $1, 2, 3, \dots, n$ in some order.
5. In a triangle ABC , D is a point on BC such that AD is the internal bisector of $\angle A$. Suppose $\angle B = 2\angle C$ and $CD = AB$. prove that $\angle A = 72^\circ$.
6. If x, y, z are sides of a triangle, prove that

$$|x^2(y - z) + y^2(z - x) + z^2(x - y)| < xyz.$$

7. Prove that the product of the first 1000 positive even integers differs from the product of the first 1000 positive odd integers by a multiple of 2001.

RMO 2002

1. In an acute triangle ABC points D, E, F are located on the sides BC, CA, AB such that

$$\frac{CD}{CE} = \frac{CA}{CB}, \frac{AE}{AF} = \frac{AB}{AC}, \frac{BF}{FD} = \frac{BC}{BA}$$

Prove that AD, BE, CF are altitudes of triangle ABC .

2. Solve for real x :

$$(x^2 + x - 2)^3 + (2x^2 - x - 1)^3 = 27(x^2 - 1)^3.$$

3. Let a, b, c be positive integers such that a divides b^2 , b divides c^2 and c divides a^2 . Prove that abc divides $(a + b + c)^7$.
4. Suppose the integers $1, 2, \dots, 10$ are split into two disjoint collections $a_1, a_2, \dots, a_5$ and $b_1, \dots, b_5$ such that $a_1 < a_2 < a_3 < a_4 < a_5, b_1 < b_2 < b_3 < b_4 < b_5$ (i) Show that the larger number in any pair $\{a_j, b_j\}$, $1 \leq j \leq 5$ is at least 6. (ii) Show that $\sum_{i=1}^5 |a_i - b_i| = 25$ for every such partition.
5. The circumference of a circle is divided into eight arcs by a convex quadrilateral $ABCD$ with four arcs lying inside the quadrilateral and the remaining four lying outside it. The lengths of the arcs lying inside the quadrilateral are denoted by p, q, r, s in counter-clockwise direction. Suppose $p + r = q + s$. Prove that $ABCD$ is cyclic.
6. Prove that for any natural number $n > 1$,

$$\frac{1}{2} < \frac{1}{n^2 + 1} + \frac{2}{n^2 + 2} + \dots + \frac{n}{n^2 + n} < \frac{1}{2} + \frac{1}{2n}.$$

7. Find all integers a, b, c, d such that

(a) $1 \leq a \leq b \leq c \leq d$;

(b) $ab + cd = a + b + c + d + 3$.

RMO 2003

1. Let ABC be a triangle in which $AB = AC$ and $\angle CAB = 90^\circ$. Suppose that M and N are points on the hypotenuse BC such that $BM^2 + CN^2 = MN^2$. Prove that $\angle MAN = 45^\circ$.
2. If n is an integer greater than 7, prove that $\binom{n}{7} - \left\lfloor \frac{n}{7} \right\rfloor$ is divisible by 7.
3. Let a, b, c be three positive real numbers such that $a + b + c = 1$. Prove that among the three numbers $a - ab, b - bc, c - ca$ there is one which is at most $\frac{1}{4}$ and there is one which is at least $\frac{2}{9}$.
4. Find the number of ordered triples (x, y, z) of non-negative integers satisfying (i) $x \leq y \leq z$ (ii) $x + y + z \leq 100$.
5. Suppose P is an interior point of a triangle ABC such that the ratios

$$\frac{d(A, BC)}{d(P, BC)}, \frac{d(B, CA)}{d(P, CA)}, \frac{d(C, AB)}{d(P, AB)}$$

are all equal. Find the common value of these ratios. $d(X, YZ)$ represents the perpendicular distance from X to the line YZ .

6. Find all real numbers a for which the equation $x^2a - 2x + 1 = 3|x|$ has exactly three distinct real solutions in x .
7. Consider the set $X = \{1, 2, \dots, 10\}$. Find two disjoint nonempty subsets A and B of X such that
 - (a) $A \cup B = X$;
 - (b) $\prod_{x \in A} x$ is divisible by $\prod_{x \in B} x$, where $\prod_{x \in C} x$ is the product of all numbers in C ;
 - (c) $\frac{\prod_{x \in A} x}{\prod_{x \in B} x}$ is as small as possible.

RMO 2004

1. Consider in the plane a circle Γ with centre O and a line l not intersecting the circle. Prove that there is a unique point Q on the perpendicular drawn from O to line l , such that for any point P on the line l , PQ represents the length of the tangent from P to the given circle.
2. Positive integers are written on all the faces of a cube, one on each. At each corner of the cube, the product of the numbers on the faces that meet at the vertex is written. The sum of the numbers written on the corners is 2004. If T denotes the sum of the numbers on all the faces, find the possible values of T .
3. Let α and β be the roots of the equation $x^2 + mx - 1 = 0$ where m is an odd integer. Let $\lambda_n = \alpha^n + \beta^n, n \geq 0$ Prove that (A) λ_n is an integer (B) $\gcd(\lambda_n, \lambda_{n+1}) = 1$.
4. Prove that the number of triples (A, B, C) where A, B, C are subsets of $\{1, 2, \dots, n\}$ such that $A \cap B \cap C = \phi$, $A \cap B \neq \phi$, $C \cap B \neq \phi$ is $7^n - 2 \cdot 6^n + 5^n$.
5. Let $ABCD$ be a quadrilateral; X and Y be the midpoints of AC and BD respectively and lines through X and Y respectively parallel to BD , AC meet in O . Let P, Q, R, S be the midpoints of AB, BC, CD, DA respectively. Prove that
(A) $APOS$ and $APXS$ have the same area (B) $APOS, BQOP, CROQ, DSOR$ have the same area.
6. Let $p_1, p_2, \dots$ be a sequence of primes such that $p_1 = 2$ and for $n \geq 1, p_{n+1}$ is the largest prime factor of $p_1 p_2 \dots p_n + 1$. Prove that $p_n \neq 5$ for any n .
7. Let x and y be positive real numbers such that $y^3 + y \leq x - x^3$. Prove that
 - (a) $y < x < 1$
 - (b) $x^2 + y^2 < 1$.

RMO 2005

1. Let ABCD be a convex quadrilateral; P,Q, R,S are the midpoints of AB, BC, CD, DA respectively such that triangles AQR, CSP are equilateral. Prove that ABCD is a rhombus. Find its angles.
2. If x, y are integers and 17 divides both $x^2 - 2xy + y^2 - 5x + 7y$ and $x^2 - 3xy + 2y^2 + x - y$, then prove that 17 divides $xy - 12x + 15y$.
3. If a, b, c are positive three real numbers such that $|a - b| \geq c, |b - c| \geq a, |c - a| \geq b$. Prove that one of a, b, c is equal to the sum of the other two.
4. Find the number of 5-digit numbers that each contains the block '15' and is divisible by 15.
5. In a triangle ABC, D is midpoint of BC . If $\angle ADB = 45^\circ$ and $\angle ACD = 30^\circ$, determine $\angle BAD$.
6. Determine all triples of positive integers (a, b, c) such that $a \leq b \leq c$ and $a + b + c + ab + bc + ca = abc + 1$.
7. Let a, b, c be three positive real numbers such that $a + b + c = 1$. Let $\lambda = \min\{a^3 + a^2bc, b^3 + b^2ac, c^3 + abc^2\}$
Prove that the roots of $x^2 + x + 4\lambda = 0$ are real.

RMO 2006

1. Let ABC be an acute-angled triangle and let D, E, F be the feet of perpendiculars from A, B, C respectively to BC, CA, AB . Let the perpendiculars from F to CB, CA, AD, BE meet them in P, Q, M, N respectively. Prove that the points P, Q, M, N are collinear.
2. If a and b are natural numbers such that $a + 13b$ is divisible by 11 and $a + 11b$ is divisible by 13, then find the least possible value of $a + b$.
3. If a, b, c are three positive real numbers, prove that $\frac{a^2+1}{b+c} + \frac{b^2+1}{c+a} + \frac{c^2+1}{a+b} \geq 3$
4. A 6×6 square is dissected into 9 rectangles by lines parallel to its sides such that all these rectangles have integer sides. Prove that there are always two congruent rectangles.
5. Let $ABCD$ be a quadrilateral in which AB is parallel to CD and perpendicular to AD ; $AB = 3CD$; and the area of the quadrilateral is 4. If a circle can be drawn touching all the four sides of the quadrilateral, find its radius.
6. Prove that there are infinitely many positive integers n such that $n(n+1)$ can be represented as a sum of two positive squares in at least two different ways. (Here $a^2 + b^2$ and $b^2 + a^2$ are considered as the same representation.)
7. Let X be the set of all positive integers greater than or equal to 8 and let $f : X \rightarrow X$ be a function such that $f(x+y) = f(xy)$ for all $x \geq 4, y \geq 4$. If $f(8) = 9$, determine $f(9)$.

RMO 2007

1. Let ABC be an acute-angled triangle; AD be the bisector of $\angle BAC$ with D on BC ; and BE be the altitude from B on AC . Show that $\angle CED > 45^\circ$. [weightage 17/100]
2. Let a, b, c be three natural numbers such that $a < b < c$ and $\gcd(c - a, c - b) = 1$. Suppose there exists an integer d such that $a + d, b + d, c + d$ form the sides of a right-angled triangle. Prove that there exist integers, l, m such that $c + d = l^2 + m^2$. [Weightage 17/100]
3. Find all pairs (a, b) of real numbers such that whenever α is a root of $x^2 + ax + b = 0$, $\alpha^2 - 2$ is also a root of the equation. [Weightage 17/100]
4. How many 6 – digit numbers are there such that:- a) The digits of each number are all from the set $\{1, 2, 3, 4, 5\}$ b) any digit that appears in the number appears at least twice? (Example: 225252 is valid while 222133 is not) [weightage 17/100]
5. A trapezium $ABCD$, in which AB is parallel to CD , is inscribed in a circle with centre O . Suppose the diagonals AC and BD of the trapezium intersect at M , and $OM = 2$. (a) If $\angle AMB$ is 60° , find, with proof, the difference between the lengths of the parallel sides. (b) If $\angle AMD$ is 60° , find, with proof, the difference between the lengths of the parallel sides. [Weightage 17/100]
6. Prove that: (a) $5 < \sqrt{5} + \sqrt[3]{5} + \sqrt[4]{5}$ (b) $8 > \sqrt{8} + \sqrt[3]{8} + \sqrt[4]{8}$ (c) $n > \sqrt{n} + \sqrt[3]{n} + \sqrt[4]{n}$ for all integers $n \geq 9$. [Weightage 16/100]

RMO 2008

1. Let ABC be an acute angled triangle; let D, F be the midpoints of BC, AB respectively. Let the perpendicular from F to AC and the perpendicular from B to AC meet in N : Prove that ND is the circumradius of ABC .
2. Prove that there exist two infinite sequences $\{a_n\}_{n \geq 1}$ and $\{b_n\}_{n \geq 1}$ of positive integers such that the following conditions hold simultaneously: (i) $0 < a_1 < a_2 < a_3 < \dots$; (ii) $a_n < b_n < a_n^2$, for all $n \geq 1$; (iii) $a_n - 1$ divides $b_n - 1$, for all $n \geq 1$ (iv) $a_n^2 - 1$ divides $b_n^2 - 1$, for all $n \geq 1$
3. Suppose a and b are real numbers such that the roots of the cubic equation $ax^3 - x^2 + bx - 1$ are positive real numbers. Prove that:

$$(i) \ 0 < 3ab \leq 1 \text{ and } (ii) \ b \geq \sqrt{3}$$

4. Find the number of all 6-digit natural numbers such that the sum of their digits is 10 and each of the digits 0, 1, 2, 3 occurs at least once in them.
5. Three nonzero real numbers a, b, c are said to be in harmonic progression if $\frac{1}{a} + \frac{1}{c} = \frac{2}{b}$. Find all three term harmonic progressions a, b, c of strictly increasing positive integers in which $a = 20$ and b divides c .
6. Find the number of all integer-sided isosceles obtuse-angled triangles with perimeter 2008.

RMO 2009

1. Let ABC be a triangle in which $AB = AC$ and let I be its in-centre. Suppose $BC = AB + AI$. Find $\angle BAC$.
2. Show that there is no integer a such that $a^2 - 3a - 19$ is divisible by 289.
3. Show that $3^{2008} + 4^{2009}$ can be written as product of two positive integers each of which is larger than 2009^{182} .
4. Find the sum of all 3-digit natural numbers which contain at least one odd digit and at least one even digit.
5. A convex polygon is such that the distance between any two vertices does not exceed 1. (i) Prove that the distance between any two points on the boundary of the polygon does not exceed 1. (ii) If X and Y are two distinct points inside the polygon, prove that there exists a point Z on the boundary of the polygon such that $XZ + YZ \leq 1$.
6. In a book with page numbers from 1 to 100 some pages are torn off. The sum of the numbers on the remaining pages is 4949. How many pages are torn off?

RMO 2010

1. Let $ABCDEF$ be a convex hexagon in which diagonals AD, BE, CF are concurrent at O . Suppose $[OAF]$ is geometric mean of $[OAB]$ and $[OEF]$ and $[OBC]$ is geometric mean of $[OAB]$ and $[OCD]$. Prove that $[OED]$ is the geometric mean of $[OCD]$ and $[OEF]$. (Here $[XYZ]$ denotes area of $\triangle XYZ$)
2. Let $P_1(x) = ax^2 - bx - c$, $P_2(x) = bx^2 - cx - a$, $P_3(x) = cx^2 - ax - b$ be three quadratic polynomials. Suppose there exists a real number α such that $P_1(\alpha) = P_2(\alpha) = P_3(\alpha)$. Prove that $a = b = c$.
3. Find the number of 4-digit numbers (in base 10) having non-zero digits and which are divisible by 4 but not by 8.
4. Find three distinct positive integers with the least possible sum such that the sum of the reciprocals of any two integers among them is an integral multiple of the reciprocal of the third integer.
5. Let ABC be a triangle in which $\angle A = 60^\circ$. Let BE and CF be the bisectors of $\angle B$ and $\angle C$ with E on AC and F on AB . Let M be the reflection of A in line EF . Prove that M lies on BC .
6. For each integer $n \geq 1$ define $a_n = \left\lceil \frac{n}{\lceil \sqrt{n} \rceil} \right\rceil$ (where $[x]$ denotes the largest integer not exceeding x , for any real number x). Find the number of all n in the set $\{1, 2, 3, \dots, 2010\}$ for which $a_n > a_{n+1}$

RMO 2011

1. Let ABC be a triangle. Let D, E, F be points respectively on the segments BC, CA, AB such that AD, BE, CF concur at the point K . Suppose $\frac{BD}{DC} = \frac{BF}{FA}$ and $\angle ADB = \angle AFC$. Prove that $\angle ABE = \angle CAD$.
2. Let $(a_1, a_2, a_3, \dots, a_{2011})$ be a permutation of the numbers $1, 2, 3, \dots, 2011$. Show that there exist two numbers j, k such that $1 \leq j < k \leq 2011$ and $|a_j - j| = |a_k - k|$.
3. A natural number n is chosen strictly between two consecutive perfect squares. The smaller of these two squares is obtained by subtracting k from n and the larger by adding l to n . Prove that $n - kl$ is a perfect square.
4. Consider a 20-sided convex polygon K , with vertices $A_1, A_2, \dots, A_{20}$ in that order. Find the number of ways in which three sides of K can be chosen so that every pair among them has at least two sides of K between them. (For example $(A_1A_2, A_4A_5, A_{11}A_{12})$ is an admissible triple while $(A_1A_2, A_4A_5, A_{19}A_{20})$ is not.
5. Let ABC be a triangle and let BB_1, CC_1 be respectively the bisectors of $\angle B, \angle C$ with B_1 on AC and C_1 on AB . Let E, F be the feet of perpendiculars drawn from A onto BB_1, CC_1 respectively. Suppose D is the point at which the incircle of ABC touches AB . Prove that $AD = EF$.
6. Find all pairs (x, y) of real numbers such that

$$16^{x^2+y} + 16^{x+y^2} = 1$$

RMO 2013

1. Let ABC be an isosceles triangle with $AB = AC$ and let Γ denote its circumcircle. A point D is on arc AB of Γ not containing C . A point E is on arc AC of Γ not containing B . If $AD = CE$ prove that BE is parallel to AD .
2. Find all triples (p, q, r) of primes such that $pq = r + 1$ and $2(p^2 + q^2) = r^2 + 1$.
3. A finite non-empty set of integers is called 3-good if the sum of its elements is divisible by 3. Find the number of 3-good subsets of $\{0, 1, 2, \dots, 9\}$.
4. In a triangle ABC , points D and E are on segments BC and AC such that $BD = 3DC$ and $AE = 4EC$. Point P is on line ED such that D is the midpoint of segment EP . Lines AP and BC intersect at point S . Find the ratio BS/SD .
5. Let a_1, b_1, c_1 be natural numbers. We define

$$a_2 = \gcd(b_1, c_1), \quad b_2 = \gcd(c_1, a_1), \quad c_2 = \gcd(a_1, b_1),$$

and

$$a_3 = \text{lcm}(b_2, c_2), \quad b_3 = \text{lcm}(c_2, a_2), \quad c_3 = \text{lcm}(a_2, b_2).$$

Show that $\gcd(b_3, c_3) = a_2$.

6. Let $P(x) = x^3 + ax^2 + b$ and $Q(x) = x^3 + bx + a$, where a and b are nonzero real numbers. Suppose that the roots of the equation $P(x) = 0$ are the reciprocals of the roots of the equation $Q(x) = 0$. Prove that a and b are integers. Find the greatest common divisor of $P(2013! + 1)$ and $Q(2013! + 1)$.

RMO 2014

1. Three positive real numbers a, b, c are such that $a^2 + 5b^2 + 4c^2 - 4ab - 4bc = 0$. Can a, b, c be the lengths of the sides of a triangle? Justify your answer.

2. The roots of the equation

$$x^3 - 3ax^2 + bx + 18c = 0$$

form a non-constant arithmetic progression and the roots of the equation

$$x^3 + bx^2 + x - c^3 = 0$$

form a non-constant geometric progression. Given that a, b, c are real numbers, find all positive integral values a and b .

3. Let ABC be an acute-angled triangle in which $\angle ABC$ is the largest angle. Let O be its circumcentre. The perpendicular bisectors of BC and AB meet AC at X and Y respectively. The internal angle bisectors of $\angle AXB$ and $\angle BYC$ meet AB and BC at D and E respectively. Prove that BO is perpendicular to AC if DE is parallel to AC .
4. A person moves in the $x - y$ plane moving along points with integer co-ordinates x and y only. When she is at a point (x, y) , she takes a step based on the following rules: (a) if $x + y$ is even she moves to either $(x + 1, y)$ or $(x + 1, y + 1)$; (b) if $x + y$ is odd she moves to either $(x, y + 1)$ or $(x + 1, y + 1)$. How many distinct paths can she take to go from $(0, 0)$ to $(8, 8)$ given that she took exactly three steps to the right ((x, y) to $(x + 1, y)$)?
5. Let a, b, c be positive real numbers such that

$$\frac{1}{1+a} + \frac{1}{1+b} + \frac{1}{1+c} \leq 1.$$

Prove that $(1 + a^2)(1 + b^2)(1 + c^2) \geq 125$. When does equality hold?

6. Let D, E, F be the points of contact of the incircle of an acute-angled triangle ABC with BC, CA, AB respectively. Let I_1, I_2, I_3 be the incentres of the triangles AFE, BDF, CED , respectively. Prove that the lines I_1D, I_2E, I_3F are concurrent.

RMO 2015

1. Let ABCD be a convex quadrilateral with $AB = a$, $BC = b$, $CD = c$ and $DA = d$. Suppose

$$a^2 + b^2 + c^2 + d^2 = ab + bc + cd + da,$$

and the area of ABCD is 60 square units. If the length of one of the diagonals is 30 units, determine the length of the other diagonal.

2. Determine the number of 3-digit numbers in base 10 having at least one 5 and at most one 3.
3. Let $P(x)$ be a non-constant polynomial whose coefficients are positive integers. If $P(n)$ divides $P(P(n) - 2015)$ for every natural number n , prove that $P(-2015) = 0$.
4. Find all three digit natural numbers of the form $(abc)_{10}$ such that $(abc)_{10}$, $(bca)_{10}$ and $(cab)_{10}$ are in geometric progression. (Here $(abc)_{10}$ is representation in base 10.)
5. Let ABC be a right-angled triangle with $\angle B = 90^\circ$ and let BD be the altitude from B on to AC. Draw $DE \perp AB$ and $DF \perp BC$. Let P, Q, R and S be respectively the incentres of triangle DFC, DBF, DEB and DAE. Suppose S, R, Q are collinear. Prove that P, Q, R, D lie on a circle.
6. Let $S = \{1, 2, \dots, n\}$ and let T be the set of all ordered triples of subsets of S, say (A_1, A_2, A_3) , such that $A_1 \cup A_2 \cup A_3 = S$. Determine, in terms of n , $\sum_{(A_1, A_2, A_3) \in T} |A_1 \cap A_2 \cap A_3|$ where $|X|$ denotes the number of elements in the set X. (For example, if $S = \{1, 2, 3\}$ and $A_1 = \{1, 2\}$, $A_2 = \{2, 3\}$, $A_3 = \{3\}$ then one of the elements of T is $(\{1, 2\}, \{2, 3\}, \{3\})$.)
7. Let x, y, z be real numbers such that $x^2 + y^2 + z^2 - 2xyz = 1$. Prove that

$$(1+x)(1+y)(1+z) \leq 4 + 4xyz.$$

8. The length of each side of a convex quadrilateral ABCD is a positive integer. If the sum of the lengths of any three sides is divisible by the length of the remaining side then prove that some two sides of the quadrilateral have the same length.

RMO 2016

1. Let ABC be a right-angled triangle with $\angle B = 90^\circ$. Let I be the incenter of ABC . Draw a line perpendicular to AI at I . Let it intersect the line CB at D . Prove that CI is perpendicular to AD and prove that $ID = \sqrt{b(b-a)}$ where $BC = a$ and $CA = b$.
2. Let a, b, c be positive real numbers such that

$$\frac{a}{1+a} + \frac{b}{1+b} + \frac{c}{1+c} = 1.$$

Prove that $abc \leq \frac{1}{8}$.

3. For any natural number n , expressed in base 10, let $S(n)$ denote the sum of all digits of n . Find all natural numbers n such that $n = 2S(n)^2$.
4. Find the number of all 6-digits numbers having exactly three odd and three even digits.
5. Let ABC be a triangle with centroid G . Let the circumcircle of triangle AGB intersect the line BC in X different from B ; and the circumcircle of triangle AGC intersect the line BC in Y different from C . Prove that G is the centroid of triangle AXY .
6. Let $(a_1, a_2, \dots)$ be a strictly increasing sequence of positive integers in arithmetic progression. Prove that there is an infinite sub-sequence of the given sequence whose terms are in a geometric progression.

Answer Key

Addition Principle

1. 35
2. 10
3. $n(n+1)/2$
4. 17

Multiplication Principle

1. 15
2. 210
3. 2^n
4. 120
5. 4500
6. 9^5
7. 374
8. 24
9. 5×10^4
10. $n(n-3)/2$
11. 24, 201110.
12. 333
13. 3136, 3612
14. 5; (5,20)(6,12)(8,8)(12,6)(20,5)

Factorials

1. 100
2. 3
3. 3
4. $50! - 1$
5. 9
6. 23

Distinct Permutations

1. 2520
2. $6! \cdot 5!$, $2 \cdot 6! \cdot 6!$
3. $8! \cdot 3!$, $7! \cdot {}^6P_3$
4. 6P_3 , $5! \cdot 3!$
5. $8 \cdot 9!$
6. $4! \cdot (3!)^2 \cdot (2!)^2$
7. $4!$, 60
8. 360×11111

Permutations with Repetitions

1. $\frac{17!}{7!6!4!}$
2. 360
3. 26^3
4. (a) 360, (b) ABELTT, TTLEBA, (c) 243, (d) TEBLEA
5. $\frac{11!}{4!4!2!}, \frac{7!}{4!2!} \cdot {}^8C_4$
6. 13

Circular Permutations

1. $7!/2$
2. (a) $7!$, (b) $6! \cdot 5$, (c) $4! \cdot {}^5P_3$
3. (a) $6! \cdot 7!$, (b) $6! \cdot 2^7$
4. 50
5. 3
6. ${}^{16}C_6 \cdot (10!)^2$
7. 30
8. 28

r-Combinations

1. ${}^6C_3 \cdot {}^8C_3$
2. 105
3. ${}^{10}C_5/2$
4. (a) ${}^{30}C_4$, (b) ${}^{30}P_4$
5. 28, ${}^{28}C_2$
6. ${}^{20}C_3, {}^{20}C_4$
7. 69
8. 390
9. (a) 56, (b) 105
10. 204, $\binom{9}{2}^2$
11. (a) 8, (b) 24, (c) 48, (d) 24, (e) 6, (f) 18
12. 9, 2592
13. $4 \cdot {}^{13}C_4, 13^4$
14. ${}^{13}C_2 \cdot {}^4C_3 \cdot {}^4C_2 \cdot {}^5P_5$
15. ${}^{20}C_{15} \cdot 2^5$
16. ${}^{16}C_1 \cdot {}^8C_2$
17. 8C_7
18. (a) $10!/3!$, (b) $10!/4$
19. $2 \cdot {}^{33}C_3 + {}^{34}C_3 + 33^2 \cdot 34$
20. ${}^{20}C_3 - {}^{10}C_3 + 5 \cdot {}^{10}C_2 = 795$
21. 156

Lattice Points and Paths

1. 120
2. (a) $T = {}^{20}C_{10}$, (b) $P = {}^5C_3 \cdot {}^{15}C_7$, (c) $Q = {}^{14}C_6 \cdot {}^6C_2$, (d) $R = {}^5C_3 \cdot {}^9C_3 \cdot {}^6C_4$, (e) $P - R$, (f) $T - (P + Q - R)$
3. ${}^5C_3 \cdot {}^7C_4, {}^{13}C_8 - ({}^5C_3 \cdot {}^7C_4)$
4. ${}^{20}C_9, 145 \cdot {}^{24}C_{11}$

Number of Integral Solutions

1. 19
2. ${}^{15}C_3$
3. ${}^{13}C_7, {}^{16}C_7$
4. ${}^{14}C_8$
5. 7
6. ${}^{19}C_4$
7. ${}^{11}C_3$
8. $2^{14}, {}^{14}C_3$

Combinatorial Identities

1. $x^4 + 8x^3y + 24x^2y^2 + 32xy^3 + 16y^4$
2. $2^7, 4^7$
3. -20
4. $2(x^5 + 10x^3y^2 + 5xy^4), 2(5x^4y + 10x^2y^3 + y^4)$
5. $\binom{15}{9}$
6. $\binom{10}{5}$
7. 35

Pascal's Triangle

1. 2^n
2. 9, 36, 84, 126, 36, 9
3. $X = 1287, Y = 3003, Z = 12870$
4. $1 - 8x + 24x^2 - 32x^3 + 16x^4$
5. 1771561
6. 15

Multinomial Theorem

1. 0, 10, -720
2. ${}^{11}C_3$
3. $3^{20}, (3^{20} - 1)/2$
4. a^4b, a^4c, b^4c
5. 3, 27, 36

Distribution Into Groups

1. (a) ${}^{13}C_3$, (b) 4^{10}
2. 472
3. $27!/(3!)^9$
4. $\frac{15!}{2!3!4!}$
5. ${}^{12}C_8 \cdot \frac{8!}{16}$
6. Hint : Distribute 4! distinct objects into 3! groups.
7. (a) 3^7 , (b) 36, (c) 15

Bijection Principle

1. 2^n
2. ${}^{27}C_5$
3. ${}^{10}C_5$
4. $n \cdot 4^{n-1}$
5. $2^{n-1} - 1$
6. $\binom{82}{20}$.

Inclusion Exclusion Principle

1. 45
2. 119, 23, 23
3. 279
4. 64
5. 68
6. 176

7. 241st

8. 64

9. 1462

Derangements

2. (a) 1, (b) 1, (c) 12, (d) 3, (e) 9

3. 135 4. 15 5. 9 6. 25 7. $9 \cdot {}^8C_4$

Exercises

1. $16!/(2)^8$

2. (a) 3472, (b) 2576, (c) 3696

3. 2×4^{1998}

4. 3×2^6

5. ${}^8C_4 \times 4$

6. $3 \times {}^9C_4$

7. 1056

8. 502

9. 81

10. 2^6

11. $\binom{14}{9} - 1$

12. 2^7

13. 24

14. 29

15. 156

16. 9900

17. 13

18. $2^{10} - 2^5$

19. 2002

20. 1440

21. $3^2 5^5$

- 22. 2252
- 23. 615
- 24. 127
- 25. 12
- 26. 9
- 27. 99

PreRMO

- 1. 33
- 2. 20
- 3. 10
- 4. 8
- 5. 630
- 6. 570
- 7. 28
- 8. 62

RMO

- 4. 320
- 6. 2^9
- 7. 385
- 9. $\frac{3^n-1}{2}$
- 11. 1996002
- 12. 4315672
- 15. 7
- 18. 479
- 20. 1405
- 21. 490
- 22. 86
- 23. 370775
- 24. 729
- 25. 520
- 27. 3^6-1
- 28. 3^7-1
- 29. 3^7-1
- 30. 3^5-1
- 33. 3780
- 35. 120
- 37. 351
- 39. No
- 41. 85
- 43. $\binom{11}{6}$
- 44. 7720
- 45. 2268
- 46. 3616
- 47. 5412
- 48. 7720
- 49. 249
- 50. $n \cdot 7^{n-1}$
- 51. $100!\binom{54}{9}$